

Survey paper

Pixels to prose: A comprehensive survey of image captioning techniques with deep learning and generative artificial intelligence

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ABSTRACT

In the era of evolving artificial intelligence, machines are increasingly emulating human-like capabilities, including visual perception and linguistic expression. Image captioning is at the intersection of these domains, enabling machines to interpret visual content and generate descriptive text. This paper provides a thorough review of image captioning techniques, from foundational methods to state-of-the-art approaches. Beginning with an exploration of primitive architectures, the review traces the evolution of image captioning models to the latest cutting-edge solutions. By dissecting the components of these architectures, the reader can gain insight into the underlying mechanisms and can select suitable approaches tailored to specific problem requirements without duplication of efforts. Furthermore, the review offers guidance on evaluating the performance of image captioning systems, highlighting key metrics for assessment. By synthesizing theoretical concepts with practical application, this paper equips readers with the knowledge needed to navigate the complex landscape of image captioning and harness its potential for diverse applications in machine learning and beyond.

1. Introduction

In this paper, we survey the existing machine-learning literature for the task of image captioning. We present a comprehensive review of the key techniques used in resolving this task. Recent works in image captioning which involve deep learning techniques are based on supervised learning techniques [57,74,78,111] which use the paired image caption data to generate image captions or descriptions. We also include unsupervised and reinforcement learning approaches in our literature review for a holistic coverage of methods for image captioning [79,82,85,87]. Since image captioning lies at the intersection of two major domains of machine learning, i.e. computer vision and natural language processing (NLP), a combined architecture is used in the majority of the image captioning models. Mainly encoder-decoder architecture is used, the encoder extracts one or multiple feature vectors from the input image. Extracting the features is an important step as these features provide the context based on which the decoder in the next generative step, generates meaningful captions. [24,112] The vanilla encoder-decoder architecture comprises Convolutional Neural Network (CNN) for visual encoding and Recurrent Neural Network (RNN) for the

language decoding part, over the years this architecture has seen considerable progress, and the quality, diversity, and richness of captions have improved considerably.

In place of simple RNN architectures, more enhanced models such as Long short-term memory (LSTM) and Gated Recurrent Units (GRU) were employed. Various attention mechanisms to capture context, spatial, and semantic relations within an input image are used [5,53,65,102]. Furthermore, Region-aware interaction learning framework [51] explicitly models the interplay between image regions and objects using dual-attention, leading to improved contextual understanding and more accurate caption generation.

Additionally, Transformers with self-attention mechanisms, employed in both encoder and decoder modules have enabled the generation of more sophisticated captions [18,55,121]. An extension of image captioning models is multi-modal architectures, which not only caption an image but also provide diverse capabilities at par with human intelligence in processing and generating multi-modal information. Like humans are able to make sense of the real world by processing the multimodal information in their surroundings and generating a language or visual response accordingly, in a similar way various vision

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language models were developed which have the capability to process inputs from multiple modalities [12]. The vision-language models are capable of various multi-modal tasks such as Visual Question Answering (VQA), image captioning, and video captioning. Thus, the shift is from task-specific architectures towards more generalized models which are capable of various multi-modal operations.

Language models like Bidirectional Encoder Representations from Transformers (BERT) [20] pre-trained on massive training datasets are used as language decoders in vision-language models [13,23,30]. Due to the utility of the task of image captioning, it has found various applications in important domains such as healthcare, autonomous vehicles, remote sensing, and helping visually impaired people retrieve information and navigate routes.

Given the wide-ranging applications and extensive research going on in this field, we conducted a systematic literature review, which aims to answer the following research questions:

- RQ1. Which image captioning models achieve superior performance for distinct caption types, including those requiring semantic and spatial alignment, and emotional expression?
- RQ2. What are the approaches in unsupervised, self-supervised, and reinforcement learning?
- RQ3. What are the primary datasets utilized for image captioning tasks, what evaluation metrics are commonly employed, how do these metrics assess caption quality, and what are the distinct characteristics of these datasets?
- RQ4. Considering the evaluation metrics employed, which image captioning models demonstrate superior performance?
- To summarize our contributions (Fig. 1) are as follows:
- We begin with reviewing the evolution of architectures of both encoder and decoder, listing down key CNN based models for visual encoding and key enhancements in language decoders.
- We survey various methodologies used to enhance the encoder-decoder architecture such as attention, transformers, scene graphs, and global versus regional features.
- We further, comprehensively review the vision-language models which have led to evolved architectures that are capable of handling multiple modalities.
- We review evaluation metrics and datasets used for evaluating the quality of captions generated through various models.

2. Search methodology

For the purpose of our review, we collected data from various databases such as Scopus, IEEE Xplore, and Web of Science, using a combination of relevant keywords such as "image captioning," "deep learning," "neural networks," and "natural language processing." Boolean operators were employed to refine the search queries and ensure the retrieval of pertinent literature. Only studies directly related to image captioning, featuring empirical evaluations or significant contributions, were considered for inclusion (Fig. 2).

Additionally, snowballing techniques were employed to identify additional relevant studies by reviewing the references of selected papers. As a result of this approach we selected 158 papers for the

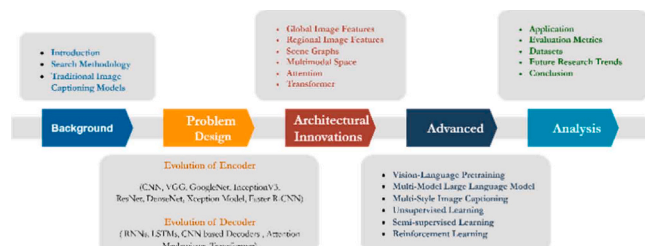


Fig. 1. Visual Outline of the Literature review paper.

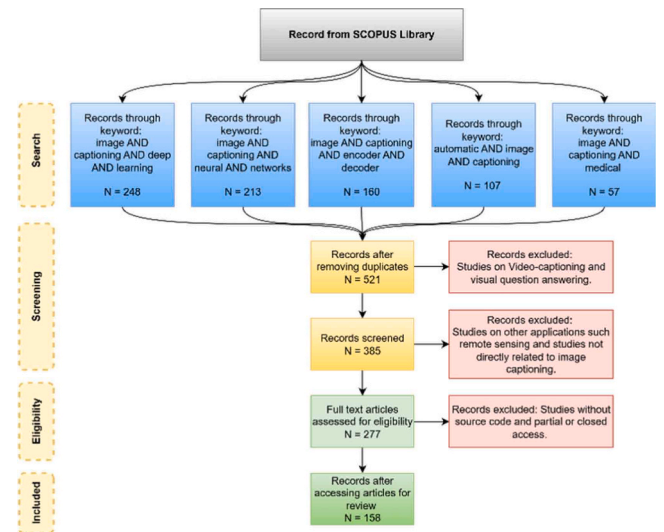


Fig. 2. Literature Shortlisting Procedure Flow Diagram.

literature review.

3. Supervised learning

3.1. Encoder-decoder architecture

Image captioning is achieved through the seamless integration of computer vision and NLP techniques, facilitating the translation of visual content into textual descriptions. Here, we review each component of the encoder-decoder architecture (Fig. 3) separately, outlining the evolution in the CNN architecture as well as the RNN architecture, in the later section, we will review how different evolved architectures with various techniques such as attention and scene graphs were employed to generate richer and coherent captions.

3.1.1. Encoder architecture evolution

In the image captioning task, the first objective is to extract feature representations from the input image and then these representations are fed to the decoder for caption generation, in between the encoder and the decoder we may have additional blocks such as attention blocks to improve the input-output alignment or capture semantics, intricate details or context within an input image.

In order to generate high-level abstractions of input images, typically a CNN architecture is used. The strong learning capabilities of a CNN architecture, derived from multiple feature extraction stages make it a suitable model for image captioning tasks, owing to this capability a CNN can automatically generate data representations. A typical CNN architecture consists of a series of convolutional and pooling layers, with one or more fully connected layers at the end. One possible substitution

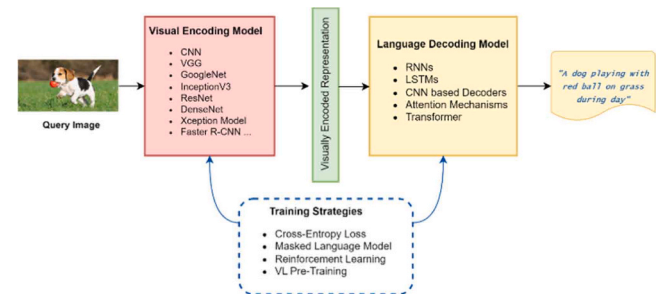


Fig. 3. A generalized view of encoder-decoder architecture with sample input and output.

for the final fully connected layer is a global average pooling layer. The CNN design incorporates many regulatory units, such as batch normalization [41] and dropout, to optimize performance. The configuration of CNN components is crucial in the construction of novel and advanced systems.

In order to enhance the capabilities of CNNs, researchers have investigated various other factors, including the utilization of different activation functions, loss functions, parameter optimization, and regularization techniques [38].

3.1.2. Few high-end CNN models used in visual encoding

By concentrating on spatial filters, the Visual Geometry Group (VGG) network brought about a novel architectural improvement in CNNs. It was found that various filter sizes capture varying degrees of granularity. In comparison to large-sized filters, smaller filters extract more fine-grained information, which enhances feature extraction performance. VGG can capture multiple levels of information in an image by using different sized filters. Notably, significant performance gains were observed when larger filters were replaced with stacked 3×3 filters. Another advantage of using small-size filters was that they reduced computational complexity by using fewer parameters. VGG uses 1×1 convolutions between convolutional layers to improve feature-map learning and control network complexity. VGG optimizes the tuning process for image analysis by placing max-pooling layers after the convolutional layer and using padding to preserve spatial resolution. But one significant disadvantage of VGG is that it has a large number of parameters (138 millions), which makes it difficult to implement on systems with limited resources [38].

GoogleNet, devised by Google researchers, aimed to achieve high accuracy while minimizing computational costs. It uses the inception block, which enhances CNNs by integrating multi-scale convolutional transformations through the split, transform, and merge paradigm. Moreover, by leveraging global average pooling instead of fully connected layers and optimizing parameter tuning, GoogleNet significantly reduced the model's parameter count from 138 million to 4 million, enhancing computational efficiency without sacrificing performance. However, GoogleNet's design also posed challenges, notably a representation bottleneck where the dimensionality of the feature space is significantly reduced in the subsequent layer. This reduction in feature space can sometimes lead to the loss of valuable information [38].

Inception-V3: One notable enhancement in Inception-V3 is the replacement of large-sized filters (5×5 and 7×7) with smaller and asymmetric filters (1×7 and 1×5). This strategic adjustment enables the network to capture spatial information more efficiently while minimizing computational costs. Additionally, Inception-V3 incorporates bottleneck layers of 1×1 convolutions before applying larger filters, effectively regulating computations and enhancing feature extraction. Another key feature of Inception-V3 is the utilization of 1×1 convolutional operation, which map the input data into smaller-dimensional spaces before applying regular (3×3 or 5×5) convolutions [38].

Residual Network (ResNet), introduces a fundamental shift in the architecture of deep neural networks by focusing on learning residual functions instead of directly approximating the underlying mapping from inputs to outputs. Traditional networks aim to learn the entire mapping function from scratch, which can become increasingly challenging as the network depth increases. However, ResNet addresses this challenge by reformulating the learning task to focus on approximating the residual mapping—the difference between the desired output and the input. In ResNet, each layer aims to learn a residual function. By stacking these residual blocks on top of each other, ResNet forms a deep network capable of learning complex mappings while mitigating issues like the vanishing gradient problem [38].

DenseNet: DenseNet stands out among convolutional neural network architectures for its innovative approach to feature connectivity and parameter sharing [37]. In contrast to ResNet, DenseNet uses a

dense connection pattern in which each layer concatenates feature maps along the channel dimension after receiving feature maps from all previous layers. This design choice promotes extensive feature reuse and facilitates enhanced information flow throughout the network, allowing each layer to directly access and build upon the features learned by all preceding layers. [33] By concatenating features instead of simply adding them, DenseNet explicitly differentiates between new information added to the network and existing information preserved from earlier layers, potentially improving the network's ability to learn complex patterns and representations. Furthermore, DenseNet's narrow layer structure reduces the number of parameters in each layer, mitigating issues like the vanishing gradient problem and enhancing computational efficiency. Additionally, DenseNet's direct access to gradients for each layer through the loss function serves as a regularizing mechanism, helping to combat over-fitting, especially in scenarios with limited training data [38].

Xception represents a significant advancement in convolutional neural network architecture, building upon the principles of depth wise separable convolution and linear residual connections. Derived from the concept of Inception, Xception takes this idea to an extreme by employing depth wise separable convolution, which splits the convolutional process into two distinct steps. Firstly, a 1×1 convolution is applied to capture cross-channel correlations, followed by a separate operation to capture spatial correlations within each output channel. The Xception architecture comprises 36 convolutional layers, forming a powerful feature extraction base for the network. By leveraging depth wise separable convolution and linear residual connections, Xception achieves state-of-the-art performance in a variety of computer vision applications while maintaining computational efficiency [15].

Faster R-CNN is endowed with region proposal modules that tells Fast R-CNN [25] where to 'look'. It accepts any size image as input and produces rectangular object proposals with an objectness score assigned to each. These proposals represent candidate regions in the image where objects might be present. The process typically involves sliding a small window (called an anchor) over the image at various locations and scales. For each anchor, the Region Proposal Network (RPN) predicts two scores: *Objectness Score* and *Bounding Box Regression*.

Objectness Score: This score indicates the likelihood of an object being present within the region defined by the anchor. **Bounding Box Regression:** This predicts adjustments to the anchor to better fit the object if it exists within the region. The objectness score helps filter out regions that are unlikely to contain objects, while the bounding box regression helps refine the proposals to more accurately fit the objects present in the image.

Once the RPN generates these proposals, they are passed on to subsequent stages of the object detection pipeline for further refinement and classification. This process significantly reduces the computational burden compared to exhaustively examining all possible regions in the image. The RPN can be trained end-to-end using backpropagation and stochastic gradient descent [69].

3.1.3. Decoder architecture evolution

In the context of Image Captioning The decoder model takes the output of the encoder and generates captions, in the earlier image captioning models RNNs were employed, but RNNs had certain drawbacks such as vanishing/exploding gradient problems and difficulty in maintaining long-range dependencies, to resolve this issue LSTMs and GRUs were used which used memory cells to capture long-range dependencies, LSTMs enhanced the caption generation process by capturing long-range dependencies, however, their recurrent architecture is complex and significant storage requirement while training using backpropagation through time, poses certain challenges.

To mitigate these issues CNN based decoders were employed which showed comparable results with LSTMs on various standard metrics. Although the issue of maintaining long-range dependencies was mitigated, the decoders still lacked the ability to generate captions which

were aligned with specific features of the image. To enhance the alignment of the input and the output attention mechanisms were being employed in the encoder-decoder architecture. Later on, with the advent of transformers, which were more computationally efficient, encoder-decoder architectures with transformer layers were introduced. Here we provide a comprehensive overview of various architectures employed as decoders for image captioning.

RNNs: To effectively model sequential data while preserving context and capturing long-range dependencies, RNNs and their variants, like LSTMs and GRUs, are commonly employed. In a vanilla RNN (Fig. 4), inputs at each time step are multiplied by weight matrices and passed through non-linear activation functions, such as hyperbolic tangent, to compute values for a layer of hidden units; these hidden layers are then used to calculate corresponding outputs. The hyperbolic tangent activation function is often chosen for its desirable properties, such as monotonicity, negative symmetry, and boundedness, making it suitable for encoding complex patterns and preserving gradient flow during training.

The distinctive aspect of RNNs lies in their ability to maintain internal state, or memory, across time steps. This memory encodes prior processing information and influences decisions at subsequent time steps, facilitating the capture of context and temporal dependencies. Backpropagation Through Time (BPTT) is the mechanism through which RNNs are trained. It extends backpropagation to sequential data by unrolling the network over time and propagating errors through each time step, enabling the network to learn from past interactions and adjust its parameters accordingly.

Advent of LSTMs: RNNs faced challenges in training due to the exploding or vanishing gradient problem, hindering their ability to effectively capture long-term dependencies in sequential data. LSTM networks were developed as a robust solution to mitigate this issue. LSTMs mitigate the vanishing/exploding gradient problem by incorporating specialized units composed of a cell, input gate, output gate, and forget gate. These components collectively enable the LSTM to regulate the information flow inside the network. At the core of the LSTM architecture are memory blocks, a set of recurrently connected sub-networks, which maintain state information over time intervals. The cell within each LSTM unit retains values over arbitrary time intervals, while the input gate controls the flow of information into the cell, the output gate regulates the output flow, and the forget gate manages the retention or discard of information stored in the cell. This design (Fig. 5) allows LSTMs to effectively capture long-term dependencies while preventing the issues associated with vanishing or exploding gradients. BPTT, a key training algorithm for LSTMs, requires storing intermediate activations and gradients over sequential time steps. This can result in significant memory consumption, particularly for long sequences, which may limit the scalability of LSTM models and increase training time and

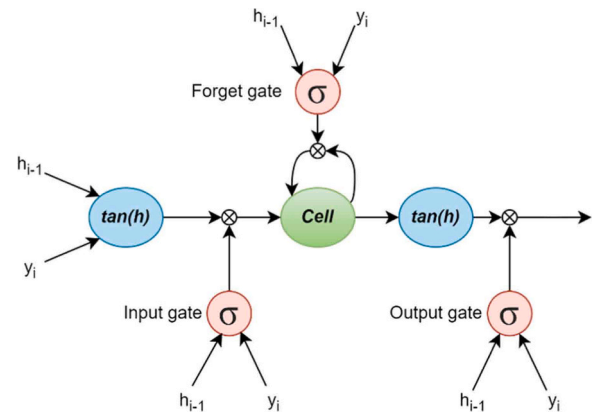


Fig. 5. A basic building block of LSTM.

resource requirements [32].

CNN based decoders: In [6], the authors introduced a CNN-based decoder for image captioning, demonstrating comparable performance to LSTM-based methods on standard evaluation metrics such as BLEU [64], CIDEr [77], SPICE [3], and ROUGE [49]. Instead of employing LSTM or GRU units, this approach utilizes masked convolutions, which are feed-forward networks. To predict a word based on previous words or their representations, they used a feed-forward deep neural network, as opposed to the RNN formulation, where the probabilistic model unfolds over time. This allows for sequential inference, with one word processed at a time. Masked convolutions are used that operate solely on past data, ensuring that future word tokens are not incorporated into the processing. Since all ground truth words are available at every time-step and there are no recurrent connections, this model can be trained in parallel for every word. This architecture provides a viable substitute for conventional recurrent-based decoders in image captioning applications by enabling effective training and inference.

3.1.4. Global image features

Global CNN features, extensively used in image encoders, capture and summarize information from the entire input, providing an overall context of an image. In the Show and Tell [78] approach, global CNN characteristics are derived from GoogleNet and subsequently inputted into the language model's initial state. In [36] the authors utilized global features from AlexNet, whereas [57] and [22] employed global features from VGG, integrating them at every time-step of the language model. The NIC model, as discussed in [79], leverages GoogleNet's last fully connected layer to activate and extract fixed-size, high-level representations of visual features. However, alternative strategies have been proposed by different researchers. For instance, the authors in [91] diverge from using the last fully connected layer output and instead utilize the output of underlying convolutional layers as the image feature vector. Additionally, in [22], a Long-Term Recurrent Convolutional Networks (LRCN) module is introduced after the initial two fully connected layers to accommodate varying-length visual input, enhancing the model's adaptability. Building upon this, authors in [88] and [99] adopt pre-trained CNNs to obtain global grid features while simultaneously learning high-level attribute representations, thereby improving caption accuracy. Furthermore, in [56], a multi-modal layer is integrated following the feature layer to fuse visual and textual representations, reinforcing their association and contributing to overall model performance. Despite these advancements, a noted drawback is that the global grid representations, although comprehensive, uniformly divide the content of an image. Consequently, this uniform fragmentation may treat relevant and non-relevant objects and regions alike, posing challenges in generating specific and accurate descriptions and leading to excessive compression of information. [59]

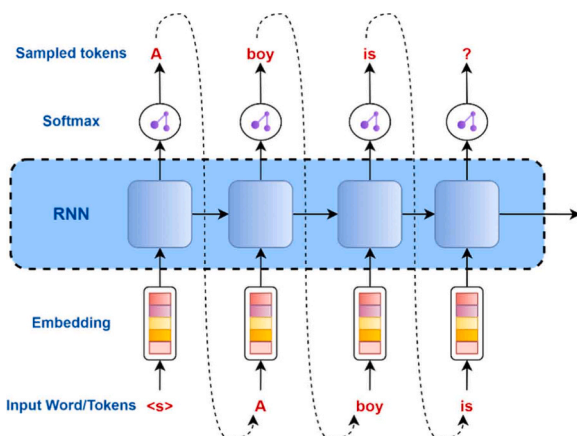


Fig. 4. Simplified RNN architecture for token generation.

3.1.5. Regional image features

In order to obtain finer understanding of the visual features, fine-grained analysis and multi-step reasoning processes are employed on the input images. This entails encoding visual features of prominent regions within images. By focusing on significant regions within an image researcher are able to delve deeper into the visual content, meaningful insights such as spatial semantics extracted from the regional image features help in generating more detailed and coherent captions. By leveraging multi-modal embedding techniques, researchers in [36] align sentence snippets with specific visual regions, thus enhancing the integration of language and image understanding. This focused analysis on particular visual areas facilitates a more nuanced comprehension of the image content. Utilizing the Faster R-CNN [69] detector as a foundation, an increasing number of methodologies are being developed to capture salient visual regions and extract high-level semantic information. Notable examples include [5] and [4]. Building upon this, authors in [19] propose a method that utilizes Regions of Interest (ROIs) and phrases within the caption to establish latent correspondences between them, thereby generating a discriminative image representation based on these matched ROIs. Additionally, in [39], Region Proposal Network (RPN) is employed to obtain object proposals, followed by a combination layer that pairs these proposals and assigns them to subject and object regions to extract contextual information. Moreover, the method presented in [93] integrates the spatial coherence of objects into a caption-generating model by combining raw visual features of overlapping object pairs to create directional pair features. These features are then optimized using learned weights, resulting in relation-aware object features that enhance the language-generating model's performance. These methods for regional feature encoding effectively address the challenge of learning fine-grained information within visual features. Beyond capturing visual representations based on global grids or salient regions, these convolutional techniques aim to identify semantic features within various regions of an image and encode them into higher-level abstractions. However, a limitation observed across many approaches is, they focus solely on entity regions, overlooking interactions between different regions. Consequently, these methods lack the ability to capture topological structure and paired relationships. [59] To enhance the granularity of visual encoding, attention mechanisms are employed, a topic which will be explored extensively in section 5.5.

3.1.6. Scene graphs

Scene graphs, operating as directed graph data structures, play a crucial role in capturing the visual semantics of images by abstracting objects and their relationships [9,108]. This abstraction facilitates the clear representation of items, attributes, and the interactions between them within a scene, aiding in the comprehension of image content. [8] Building upon this concept, the Tensor Product Scene-Graph-Triplet Representation (TPsgtR), proposed by [74], introduces a novel technique for caption generation by employing neural-symbolic encoding of scene graphs. Rather than depending on the model to construct every potential combination, neuro-symbolic embeddings of scene-graphs facilitate embedding the identified relationships among various regions of the image into tangible representations. Similarly, in [10,11] the authors introduced the Abstract Scene Graph (ASG) structure to address the gap in image captioning models, which often are unable to generate diverse descriptions according to user intentions.

The ASG structure regulates the amount of detail in generated descriptions and allows for the fine-grained depiction of user intentions. Moreover, in [92] the authors proposed scene graph captioner which tackles the main challenge; that is, to construct scene graphs that capture the rich annotations of objects, attributes, and relationships within one image. The authors suggest a scene graph captioner that produces logical descriptions by taking into account object interactions and localizations. To overcome the challenges caused by the lack of paired datasets, the researchers in [26] introduce a scene graph-based method

for unpaired image captioning. The model is trained first on the text modality, and then it uses an unsupervised feature alignment technique to align scene graphs between sentences and images.

Additionally, in [94,110] the authors incorporate human-like biases into image captioning through a scene graph auto-encoder. They generate scene graphs for both image and sentences. In the textual domain, a Scene Graph Auto-Encoder (SGAE) is employed to learn a dictionary that aids in reconstructing sentences. In the vision-language domain (Fig. 6), this shared dictionary guides the encoder-decoder model in generating captions from images. By leveraging the scene graph representation and the shared dictionary, the inherent biases present in language are effectively transferred across both visual and textual domains, facilitating more human-like and contextually rich captions.

Attention

In image captioning tasks, the attention mechanism typically unfolds in two core steps: first, determining the distribution of attention over the input features, and second, generating a context vector based on that distribution. Initially, the model processes the input features, which are transformed into a set of keys (K)—these encode salient aspects of the source input, such as regions within an image. Alongside, a task-relevant vector called the query q is introduced. The interaction between keys and query is quantified using a score function f , producing energy scores e that reflect how strongly each key relates to the query. This scoring process guides the model in identifying which parts of the input are most relevant for generating the output. [62] Subsequently, a normalization function g converts these raw energy scores into attention weights, effectively turning them into a probability distribution. To compute the final context vector, another feature representation called values (V)—aligned with the keys—is utilized. Each value corresponds directly to an element in K (Fig. 7).

By weighting these values according to the computed attention scores, the model derives a single, weighted context vector c . This vector emphasizes features with higher relevance, as indicated by the attention weights, thereby enhancing the model's ability to focus selectively and generate accurate textual descriptions from visual input [62].

The integration of attention mechanisms has notably enhanced image captioning quality by effectively capturing image semantics. By dynamically focusing on specific regions of the image, attention mechanisms facilitate the generation of contextually rich captions. In the earlier Show and Tell model [78] the visual features were extracted from a CNN enhanced with batch normalization and were fed into an LSTM decoder. An advancement in this trajectory was the Show, Attend and Tell model [91], in which attention mechanism was applied on extracted visual features. These attended visual features were concatenated with word embeddings generated from the caption sentence and were passed on to the LSTM decoder. The use of attention mechanism enables the decoder to attend over specific regions in the image and word embeddings simultaneously. This helps in learning patterns between words and visual features in a shared embedded space. Top-down attention mechanisms use the associated caption in an image caption pair to selectively focus on specific parts of the image, whereas bottom-up attention mechanism works by first extracting salient regions from the input image and then selectively attending to these regions to generate a caption, [5] combines both these attention mechanisms to generate

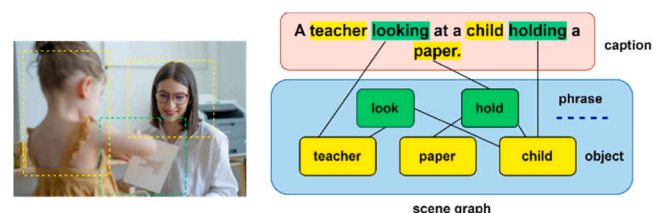


Fig. 6. A general scene generation process [48].

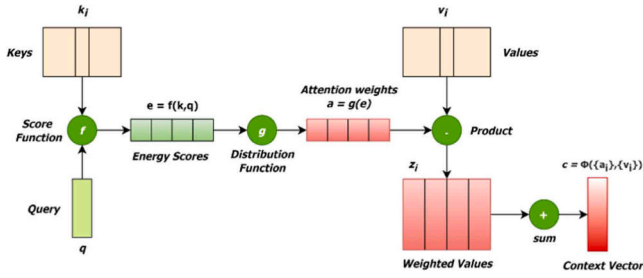


Fig. 7. Unified Attention Model's Generic Architecture.

fine-grained image captions with deeper understanding. (Fig. 8) Unlike conventional approaches where visual attention is active for every word generation step, adaptive attention model [53] makes a dynamic decision about whether to focus on the image or use a visual sentinel.

The model introduces a concept of a visual sentinel, which serves as a proxy for non-visual or less visually dependent words in the caption. When the model decides not to attend to the image, it focuses its attention on this visual sentinel instead, which is stored in the decoder memory. This helps the model to generate words that are less reliant on visual context, such as articles ("the", "a"), prepositions, and other common language constructs. An attention mechanism that simulates the pairwise interactions between the RNN state, the image regions, and the caption words is proposed by areas of attention [65].

The region-state interaction term enables the model to emphasize and suppress image regions based on their appearance and the current state. Additionally, the interactions between RNN state and words contribute to enhancing the model's ability to maintain temporal coherence in the generated word sequence, ensuring consistency as the model recursively conditions on all previous words. An attention module creates weighted average vectors based on the similarity scores between the key and query. These weighted average vectors may be irrelevant or misleading, to tackle this task attention on attention mechanism was proposed [34]. The relevancy of the query and the attention outcome is determined by the Attention on Attention mechanism.

It was observed that intra-modality information within each modality is complementary to inter-modality information. For instance, in the context of image feature extraction, each region of an image should gather insights not just from the words or phrases associated in the caption, but also from interconnected image regions. Similarly, for captions, gaining a deeper comprehension involves inferring meaning from additional words or phrases. Inter and intra-modality attention mechanisms are used in many vision-language tasks such as Visual Question Answering (VQA) to capture high-level interactions between vision and language domains.

In [66] a dynamic inter and intra-modality attention flow is proposed for multi-modal fusion. To deal with the problem of internal covariate shift in self-attention mechanisms [28] proposed normalized

self-attention which performs normalization of hidden activations on self-attention to fix their distribution. In order to capture the intra-modal relation within an image in terms of geometry, geometry aware self-attention mechanism is proposed. Furthermore, the authors in [83] propose a context-aware attention mechanism that leverages previously attended visual content to guide attention selection using an additional interpolation gate which fuses the current normalized attention weight with previously produced attention weights (Table 1).

3.1.7. Early transformer models

The Transformer architecture replaces traditional recurrent and convolutional neural networks by relying entirely on attention mechanisms to model dependencies between inputs and outputs, regardless of their distance in the sequence. [76] At the heart of this architecture lie two foundational layers: The Multi-Head Attention mechanism and the Position-wise Feed-Forward Network (FFN). These are organized within a layered encoder-decoder framework, which forms the basis for the Transformer's powerful sequence modeling capabilities. In the encoder, layers of self-attention are interleaved with feed-forward networks. This layered setup enables the model to learn intricate associations among tokens in the input sequence without any recurrence.

The decoder similarly utilizes self-attention for current output tokens, but it also incorporates cross-attention layers that interact with the encoder's output, facilitating the generation of coherent and context-aware outputs. The position-wise FFN is essentially a dense neural network applied identically across all token positions. While it processes each token independently, it preserves the positional structure of the sequence through prior encoding, allowing the model to retain contextual cues effectively.

A key innovation of the Transformer is its multi-head attention mechanism, which allows the model to attend to information from multiple perspectives simultaneously. Each "head" captures different features or relationships within the input sequence, similar to how different channels operate in convolutional networks. This multi-view approach significantly enhances the model's expressiveness. Additionally, the computational cost of self-attention remains stable even when modeling long-range interactions, making the architecture highly efficient and well-suited for parallel processing [62].

In transformer-based image captioning models, both the encoder and the decoder employ transformer-like layers and attention mechanisms. In order to obtain the visual representations, mainly two approaches are used, using a pre-trained object detector or a vision-transformer. Vision transformers are becoming compelling alternatives to the pre-trained object detectors as transformer-like architectures can be directly applied to image patches thus limiting or eliminating the need for a convolutional operator.

A typical encoder-decoder architecture with transformer is composed of a self-attention and a feed-forward layer at each encoder layer, while a self-attentive, a cross-attentive, and a feed-forward layer make up each decoder layer. [17] Architectural enhancements can be made by using the spatial information, integrating semantics and geometrical relationships between objects. Researchers have developed a number of improvements to produce more detailed captions, which we go over in this section.

The authors of [30,109] suggest an object relation transformer that uses geometric attention to integrate the spatial arrangement among identified objects in the input. Typical attention mechanisms tend to find it difficult to identify visual signals. The authors in [46] proposed the idea of using semantic attributes from language to better explicitly identify the visual signals. They propose a framework which utilizes the vision and language modalities simultaneously to close the semantic gap. Information flow from the visual and language modalities is guided using a Gated Bilateral Controller (GBC). In various attempts to enhance the typical transformer-based encoder-decoder architecture, the authors in [18] use a different design to link the encoder and the decoder architecture. The authors opt for a novel approach by feeding the decoder

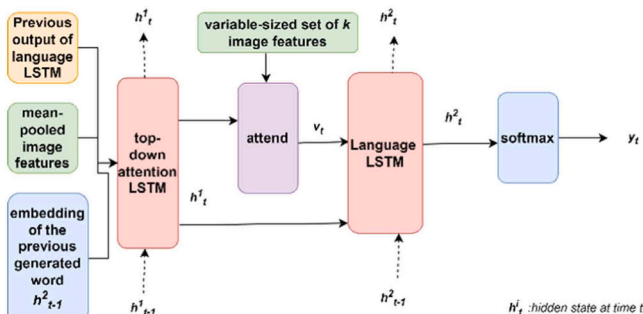


Fig. 8. Overview of bottom-up and top-down attention model [5].

Table 1

Listing the highlights and limitations of the discussed attention mechanisms.

Attention Mechanism	Highlights	Limitations
Show, attend and tell [107]	- Significant work that introduced dynamic, context-based attention to image captioning, going beyond, static encodings based on the entire image. - Demonstrated that visual grounding components of generated captions to relevant image components improves interpretability.	- Difficult to precisely localize small or irregularly shaped objects owing to the attention mechanism which operates in a coarse grid of convolutional features. - Attention mechanism attends to abstract features instead of distinct object instances, thus this mechanism lacks an object-level understanding. This might lead to confusion in cluttered scenes.
Bottom-up and Top-down attention [5]	- Bottom-up component extracts a diverse set of visual features from the input image - Top-down mechanism employs attention to focus on specific regions of the image that are most relevant to the generation of each word in the caption - Capable of generating semantically rich captions about specific objects and their attributes by attending to high-level concepts.	- Non-adaptive spatial attention decoder does not know when to use visual signals and when to use language models for generating the next word in the sentence. - It is forced to consult visual features at every time step, even when generating non-visual words (e.g., "a", "the", "of"), which can introduce noise.
Adaptive attention [63, 118]	- Adaptive attention mechanism employed by decoder which doesn't require visual attention to generate non-visual words - Better visual-to-text alignment achieved, comparable with human intuition [63] - Gating mechanism significantly reduces noise from irrelevant visual features.	- The model lacks explicit mechanisms to discriminate between words that are truly non-visual and those that are technically visual but have a high correlation with other words [63] - Spatial attention maps derived from coarse feature maps result in loss of spatial resolution and detail for smaller objects [63]
Intramodal and Intermodal attention [78]	- This mechanism enables the information flow within and across different modalities thus it is better equipped to answer questions which require both visual comparison and linguistic understanding. - These types of models are useful in various multi modal vision-language tasks such as VQA and image-text-retrieval tasks	Intra-modal attention can strengthen the correlations between all visual objects, including the non-significant background clutter. This noise amplification might overwhelm the inter-modal attention mechanism thus making it difficult to isolate the most relevant visual signal.
Context-aware attention [99]	- Models the relationship between various regions of interests in an image using a graph neural network thus moving beyond simple object co-occurrence to explicit semantic and spatial relationships. - Previously attended visual content is memorized and used to guide attention, to prevent attending to the same regions at different timesteps.	- Model parameters grow exponentially upon increasing the hyperparameters [99] due to pairwise interactions modeled in graph neural network. - The definition of nodes and the criteria for creating edges are significant hyperparameters in initial graph construction, thus a poorly constructed graph might lead the graph neural network to learn spurious

Table 1 (continued)

Attention Mechanism	Highlights	Limitations
Normalized self-attention [32]	- Model attends to more diverse and salient regions, such as the head region of a zebra and the snow region, when generating relation words like "standing" and "riding" [99] - Compensates for the lack of transformers to model geometric relations by integrating geometric inductive bias in self-attention mechanism. - Uses geometry-aware self-attention for obtaining geometric relationships between objects in an input image thus explicitly forcing the model to consider spatial context, overcoming the difficulty of standard self-attention mechanism. - Uses normalized self-attention to eliminate the problem of internal-covariate shift thus the output distribution of attention layer remains stable and well-behaved.	- relations which further leads to poor caption quality. - While geometry-aware self-attention captures relative positions, it treats all object relationships as equally important. It lacks a hierarchical or semantic understanding of these relationships. - The self-attention mechanism has a computational cost that scales quadratically with the number of image regions/objects, making it computationally expensive for images with many objects.
Areas of Attention [77]	- Novel attention mechanism which uses three pairwise interactions between RNN state, image regions, and word embedding vectors. These three pairwise attention signals allow the model to weigh different sources of information independently. - Thus, if dynamic attention is ambiguous then the other attention signal based on static semantic grounding can be used to select the correct noun.	As the model becomes more complex with additional interactions, interpreting the attention mechanism and understanding how specific interactions contribute to the model's decision-making process becomes more challenging. This can reduce the model's transparency and interpretability.
Attention on Attention [38]	- After an initial attention pass retrieves a visual context-vector the AoA module reevaluates this vector against the original query, enhancing contextual awareness. - This refining process ensures that the attention module returns a relevant output and only useful information is fed to the decoder by suppressing noise and amplifying the most relevant signal.	Stacking of attention mechanism generates a deeper computational graph, thus introducing optimization challenges such as vanishing or exploding gradients requiring more careful application of normalization layers and hyperparameter tuning.

with the encoder output in a meshed manner. Specifically, they leverage both low-level and high-level visual relationships from the encoder, enhancing the decoder's input and potentially improving performance.

The Dual-Level Collaborative Transformer (DLCT) [55] incorporates both global grid features and local region features to generate seamless captions. In order to understand both the individual characteristics of each feature type and their spatial relations, Comprehensive relation attention scheme is used. A geometric alignment graph is constructed to guide semantic alignment between region and grid features, facilitating

precise interaction. To enhance the quality of captions efforts were also directed towards improving the language decoder, such as employing knowledge graphs to improve the contextuality of the generated captions. In [100] the authors leverage knowledge graphs in transformers to generate image captions. We summarize the transformer-based architectures reviewed in this section in Table 2.

3.1.8. Multimodal fusion and alignment strategies

Learning joint feature spaces for images and their captions by projecting image features and sentence features into a common space, is useful in image search and ranking image captions. A language encoder, a vision module, a multi-modal space component, and a language decoder are the main architectural elements of a typical multi-modal space-based system [35,122].

The vision module uses a deep convolutional neural network to identify visual properties in images by extracting features from them. The language encoder, meantime, is occupied with obtaining word characteristics and generating dense embeddings for every word. These embeddings are then fed into recurrent layers, allowing it to gradually capture semantic context. Subsequently, the multi-modal space component plays a crucial role in aligning the extracted image features with the word features and mapping them into a common space. This alignment facilitates the integration of visual and textual information. Finally, the language decoder utilizes this aligned representation to generate captions [35].

Expanding on this concept, researchers in [45] propose a multi-modal image captioning model which retrieves relevant images given a sentence query, and vice-versa. They demonstrated that considering both the global content of images and sentences, as well as the specific details within them, can greatly enhance performance of

image-sentence retrieval tasks. By taking into account both the global and the finer details, the system becomes more effective at understanding and accurately retrieving relevant information.

In this direction, they proposed a model which embeds the fragments of images and fragments of sentences into a shared space rather than mapping entire images or sentences into a common embedding space. They use two objectives to design their objective function, Global ranking objective ensures that an image-caption pair is consistent with the corresponding ground truth, and all sentence fragment appearances in the visual domain are learned via Fragment Alignment.

In a similar vein, the authors in [50] employs two Multi-Modal Log Bilinear Models i.e. Modality-Biased Log-Bilinear Model (MLBL-B) and Factored 3-way Log-Bilinear Model (MLBL-F) which are extensions of the log bilinear model. Modality-Biased Log-Bilinear Model is used to predict the next word representation with a bias introduced by the features extracted from the image. MLBL-F model incorporates modality conditioning by gating the word representation matrix using the image features. The joint image-text feature learning mechanism involves using small feature maps learned using k-means to learn convolutional networks instead of using original images. Feature responses from the fully connected layer with ReLU activation are used either for additive biasing or gating in the MLBL-B and MLBL-F models, respectively.

Building upon their previous efforts, the researchers extended their work by integrating Structured Content Neural Language Models (SC-NLMs) [51] to learn joint image-sentence embeddings. This approach involved utilizing LSTM networks for sentence encoding and SC-NLMs for caption generation. The SC-NLMs were particularly effective in extracting the structural components of sentences, thereby enhancing the model's understanding of the content. This integration led to a notable improvement in caption realism compared to earlier

Table 2
Comparison of Transformer Based Image Captioning Models (Dataset: MS COCO with Karpathy Split).

Transformer Based Model	Highlights	Limitations	Evaluation Metric			
			BLEU-4	METEOR	CIDEr	SPICE
Object Relation Transformer [34]	The standard transformers rely solely on positional encodings, but object Relation Transformer modifies attention scores based on geometric relationships such as relative position, size; between different pairs of objects. This enables learning of structural and spatial semantics more effective.	The model's performance is heavily reliant on the quality of its upstream object detector. It struggles with long-tail phenomena, where rare objects or uncommon relationships are underrepresented in the training data, leading to an inability to generalize. This results in captions that are often generic and fail to capture nuanced, fine-grained details present in the image.	38.6	28.7	128.3	22.6
ETA [55]	Uses entangled attention to utilize the visual and language information simultaneously. Visual modality guides attention to the language modality and language modality guides attention to the visual modality. This bidirectional fusion enables a more effective and holistic alignment between the two modalities.	In such a tightly coupled system there is a potential risk that during training one modality might begin to dominate the other. This can pose challenges for the model to leverage the unique information present in each modality. Moreover, this entangled attention mechanism poses challenges in interpretability as well, since it becomes difficult to discern how a specific visual influences a particular generated word.	39.3	28.8	126.6	22.7
Meshed Memory Transformer [19]	Meshed connectivity allows higher layers of the encoder to directly access representations from lower layers. This creates a rich multi-scale visual encoding. A significant advantage of meshed connectivity is the ability to generate captions for images and scenes which are not well-represented in training data.	In the M2 Transformer, complicated non-linear dependencies affect how much a single region contributes to the output, in contrast to recurrent based captioning models where attention weights over regions are used which can be easily recovered. It is difficult to properly discern the relevance of particular sections in the generated captions due to this intricacy.	39.1	29.2	131.2	22.6
Dual Level Collaborative Transformer [65]	Uses cross-attention to merge grid and regional features to generate semantically rich captions	DLCT requires two separate feature extractor upstream models, any biases or errors from these initial feature extractors might be propagated and potentially amplified by DLCT, thus degrading the quality of output generated.	39.8	29.5	133.8	23.0
Transformer with Knowledge Graph [119]	Leveraging a knowledge graph during training results in a more robust model due to its exposure to a vast amount of commonsense and factual knowledge.	The model's performance is fundamentally capped by the quality, coverage, and accuracy of the external knowledge graph. Any noise, bias, or incompleteness in the knowledge graph will be directly inherited and potentially amplified by the captioning model, leading to factually incorrect or biased descriptions.	33.72	27.45	110.30	20.64

methodologies.

Furthermore, the authors in [12] use RNN for bi-directional image-sentence retrieval, incorporating a long-term visual memory to retain information about previously generated words. The primary objective is to maximize the likelihood of a word and its associated visual features given the context of previous words and their corresponding visual representations. In essence, the network compares its visual memory, representing its prior statements, with the current visual input to predict the next word in the sequence.

Additionally, an image captioning Multi-Modal RNN model (m-RNN) is presented in [67]. Each time frame of the suggested m-RNN model consists of the following five layers: two word embedding layers, a recurrent layer, a multi-modal layer, and a SoftMax layer. The syntactic and semantic meanings of the words are encoded by the two word embedding layers. The word-embedding layer, the recurrent layer, and the image representation are the three inputs of the multi-modal layer. To obtain the activation of the multi-modal layer, the activations of the three layers are mapped to the same multi-modal feature space and then added together.

With the emergence of transformers, many architectures utilize transformer-based fusion strategies. A significant strategy is to use a multimodal encoder where both image region representations and linguistic representations are processed together through multiple layers of self-attention and cross-attention [148]. The contemporary research also focuses on the direction and depth of information flow between different modalities, thus the inclination is towards bidirectional multimodal fusion and multi-stage fusion.

In bidirectional multimodal fusion there is a two-way flow of information, BMFNet [152] uses bidirectional channel-level multimodal feature fusion module (CMFF) in the encoder. In the decoder cross-attention mechanism is used for multistage fusion.

Image captioning task requires precise alignment of text and visual modalities, various attention mechanisms have shown to enhance alignment. Attention on Attention mechanism [38] uses gating layer to the attention output, the gating layer helps in gaining the contextuality of the attended image region with the generated caption.

Another approach to improve alignment is iterative cross-modal alignment. The significance of this approach is that the refinement is done in a step by step manner, thus improving semantic enhancement gradually. The authors in [153] employed iterative alignment using cross-attention for more accurate cross-modal representation fusion.

3.1.9. Decoder-only architectures

Although encoder-decoder architectures were effective in image captioning tasks, they posed challenges in the flow of information from the visual to the linguistic domain. The remarkable success of LLMs has opened up new avenues in zero-shot capabilities, training efficiency and accuracy. The architecture of decoder-only models helps in resolving the bottleneck of encoder-decoder architectures by streamlining the information flow process. The key idea is to condition a pre-trained language model directly on visual information, methodology called visual prefixing. The visual features extracted from vision models such as a CNN or a vision transformer (ViT) are converted into a sequence of embeddings which act as a 'head start' for the language model.

Now with richer contextual understanding the text generation is more grounded in visual content. ClipCap's architecture [70] combines CLIP's image encoder with lightweight linear mapping to convert image embeddings into a visual prefix for GPT-2. The key advantage is its strong performance for image captioning with minimal fine-tuning. Frozen [91] architecture is also based on the similar theme employing GPT-2 [80] as a decoder using visual prefix tokens prepended to the decoder's textual input.

Flamingo [2] developed by DeepMind is built on top of a frozen LLM decoder (such as chinchilla). Visual information is injected using Perceiver Resampler layers and then fed into the LLM's cross-attention blocks. The researchers of SimVLM [102] utilized a unified

transformer which is trained to predict the next word based on the given prefix of image and linguistic input, thus allowing direct decoding.

BLIP-2 (*Bootstrapping Language-Image Pretraining*) [123] uses a frozen visual encoder and a lightweight transformer Q-Former (Querying Transformer). Q-Former extracts the most significant image features from the frozen visual encoder and feeds them into a frozen LLM. In a two-step training process, Q-Former first learns the vision-language representations and then generates captions based on those learned representations. Using the frozen models reduces the resource intensiveness along with utilizing the vast knowledge base of LLMs. Despite significant performance a key bottleneck of decoder-only architectures is their limiting ability to perceive unidirectional context. That is, it can only look at previous tokens in a sequence to predict the next token (Table 3).

3.2. Self-supervised learning

3.2.1. Vision-language pre-training

With the rapid progress in the world of AI, we expect artificial intelligence models to learn about the world just like humans do. Humans use information from multiple channels such as vision and text and extract meaningful information from them for their learning. Based on the similar concept we have seen a new training paradigm Vision-language pre-training which mirrors human cognition's ability to synthesize information from diverse channels such as vision and language. This novel training paradigm enables the seamless integration of visual and textual data, facilitating tasks like image captioning, VQA, and image-text retrieval. We will be looking at the task of image captioning from the lens of vision-language pre-training (VLP). We have looked at image captioning models which were task-specific. They were mainly based on sequence to sequence modeling based on machine translation framework. VLP models are multi-modal in nature and can perform many vision language tasks such as image captioning, VQA, and video captioning seamlessly.

The introduction of BERT [20] [50] in NLP marked a pivotal moment for vision-language problems, driving a transition towards transformer-based multi-modal fusion models. (Fig. 9) These models are pre-trained on extensive image-text datasets, encompassing up to 4 million images and roughly 10 million image-text pairs. Examples include UNITER [12] and OSCAR [47]. More recently, the landscape has been reshaped by large-scale multi-modal foundation models like SimVLM, Florence [97,116], Flamingo [2], CoCa [96], and GIT [81]. These models have been pre-trained on an even larger scale, with over 12 billion image and text pairs. Their extensive pre-training enables adaptation to a broad spectrum of downstream tasks. [23]

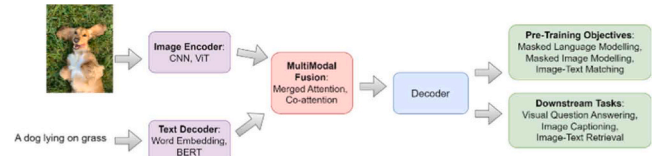
Vision Language Pre-Training Methods Within the realm of image-text tasks explored in scholarly literature, our investigation centers on the application of VLP to image captioning. The growing interest in VLP is due to two key factors. Firstly, the remarkable success achieved in language model pre-training has catalyzed interest in extending similar methodologies to multi-modal data. Secondly, the convergence of NLP and computer vision architectures has opened up fresh directions for interdisciplinary research.

In practice, VLP entails training models on large sets of image-caption pairs, allowing them to capture diverse multi-modal information. These pre-trained representations are helpful for a range of downstream tasks, underscoring the significance of VLP across various vision-language tasks. VLP methods can be broadly categorized into two main types: dual encoder and fusion encoder architectures. In dual encoder models, images and text are independently encoded, with modality interaction limited to a straightforward cosine similarity calculation between the resulting feature vectors. Conversely, fusion encoder architectures employ additional Transformer layers alongside separate image and text encoders to facilitate more intricate interactions between image and text representations. Notable examples of fusion encoder models include UNITER, VinVL [117], SimVLM [102], and METER [25].

Table 3

Summarization of various architectural innovations as discussed, along with their strengths and weaknesses.

Feature/ Technique	Strength	Weakness
Global Image Features	Simple, Compact, provides overall context. This approach is computationally inexpensive and fast, the resulting feature vector summarizes the entire scene providing a good overall context. Moreover, the image condensed as a single vector is memory efficient and easy to feed to a subsequent decoder.	Treats significant and non-significant information alike, poses a challenge in generating more fine-grained and accurate captions. Difficult to generate captions that describe specific objects, attributes or their interactions, generating very generic captions.
Regional Image Features	Localized, captures salient information grounded in key object instances present in the image. By identifying the individual objects, it provides the necessary groundwork for more advanced models like attention and scene graphs to deduce more intricate relationships within a scene.	Focuses only on entity regions, doesn't capture the topological structure and object relationships. The quality of the final caption depends on the accuracy of the underlying object detector, if the detector fails to identify any object it cannot be included in the caption.
Multimodal Fusion and Alignment	Helps in bidirectional image-text retrieval, enhanced contextual understanding owing to enhanced fusion and alignment strategies. This also enables zero-shot capabilities as the models understand concepts at a more abstract level. Another advantage is bi-directional image-text retrieval.	Poor alignment or fusion of visual and textual embeddings may result in inaccurate captions, the models might learn spurious correlations leading to hallucinated captions especially with unclean and noisy web-data. Models trained on one domain might perform poorly on another specialized domain like medical or remote-sensing, this adaption poses a challenge.
Scene Graphs	Captures object-attributes relationships and spatial arrangements in an image, provides topological representation of a scene, thus generating more compositionally accurate captions. The structured output can also be used for VQA which requires reasoning about a scene's content.	Limited representation of dense interactions, Complex hierarchical interactions may pose challenges. A predefined and finite vocabulary of object and predicate classes poses a constraint. They cannot describe concepts, actions, or relationships that are not in this vocabulary
Attention	Enables selective focus on relevant features, facilitates alignment between image regions and corresponding words in generated captions, incorporates contextual information, enables adaptive focus on different image regions based on their relevance to the current word being generated, ability to handle diverse images with varying compositions, contents, and complexities	Computational overhead, the computational cost scales quadratically with the number of features. Sometimes the model might focus on irrelevant information leading to hallucinated outcome. While it is good focusing on localized details it might pose challenge in overall scene context because it processes image region by region.
Transformer	Allows parallel processing of input, effective in capturing long-range dependencies, employs attention mechanism to focus on relevant regions when generating captions, scalable, transfer learning.	Transformers have a weaker "inductive bias" than CNNs (which are built to understand pixel locality). Consequently, they require enormous amounts of pre-training data to learn these patterns from scratch.

**Fig. 9.** General Framework for Transformer-based vision-language model [23].

While dual encoder models excel in tasks such as image retrieval, fusion encoder architectures demonstrate superior performance in image captioning and VQA tasks. This review will focus primarily on fusion encoder methods within the context of VLP approaches for image captioning [27].

Fusion encoder methods: Fusion encoder comprises multiple layers of transformer blocks which efficiently combine fine-grained features from images-text pairs. First the image features and textual words are encoded into a common embedded space. This process ensures that information from both modalities is represented in a unified format. Then a transformer module is applied to this common embedding to learn contextualized embeddings for each region and word through various pre-trainings tasks. E.g. UNITER, SimVLM, METER. Fusion encoder methods further have two categorizations - Pre-trained object detector based VLP modes and end-to-end pre-training methods.

Pre-trained Object Detector based Methods - These techniques extract visual characteristics using pre-trained object detectors, and the multi-modal fusion module generates multi-modal representations using a variety of attention processes, including co-attention and merged attention. Cross-attention is later utilized to enable cross-modality collaborative learning; in co-attention, distinct transformers are used to extract region-based visual and textual information. E.g. ViLBERT [62], LXMERT [90]. In merged attention visual and textual features are fed into a transformer block after being concatenated. E.g. UNITER [13], OSCAR [56].

End-to-End pre-training methods - In end-to-end pre-training methods various methods are used to extract visual features, mainly convolutional neural networks and Vision transformers are used to extract visual features. The model gradients are then back-propagated to the vision backbone for an end-to-end training. Various pre-training tasks are used for end-to-end training such as Masked language modeling and image text-matching. Pixel-BERT [40] uses CNN for visual embedding learning and deep multi-modal transformers to jointly learn visual and language embeddings. SOHO [39] extracts visual features using CNN and then aggregates these features into a visual dictionary. Later a transformer is used to fuse visual and language modalities. Vision transformers are used in various VLP models to encode visual information. Some significant examples include, ViLT [48] which directly feeds concatenated image and word embeddings into a transformer.

3.2.2. Case studies of various VLP architectures

Unified Architecture Various vision-language tasks can be accomplished by unifying several image and text related tasks as a single objective, such as text-generation or language modeling, the authors in [13] introduce VL-T5 and VL-BART which unifies various image-text tasks with a unified text-generation objective, they use pre-trained language models for text generation and eliminate the need for task-specific hand-crafted architectures and objectives. This process seeks to generate text that maintains coherence and meaning irrespective of the input format—image, audio, or video. The focus lies on generating text that accurately reflects the information conveyed by the various input modalities, maintaining relevance and coherence throughout the generated content [13].

Frozen: This method uses the pre-trained large language model for various multi-modal vision language tasks. The architecture of frozen consists of a visual encoder which takes raw image as input, the vision encoder's outputs are linearly mapped and reshaped to generate a

sequence of continuous embeddings, so that these visual representations can be fed into the pre-trained language models. Frozen is different than other models in a way that to apply this method to various vision language tasks does not need updating the weights of the language transformer as activations move through the model, the system becomes more adept at handling tasks that involve both vision and language, gradually adapting and enhancing its performance on multi-modal challenges. [75].

Video Language Model: The Video-Language Model (VLM) simplifies pre-training by using a single BERT encoder, focusing on token masking without task-specific alignment. VLM also demonstrates the ability to fine-tune a single encoder for various tasks using task-specific attention masks, improving task performance [89,105].

OFA represents a significant stride towards unified architectures for diverse vision and language tasks, including vision-only, language-only, and combined tasks. It serves as a task-agnostic and modality-agnostic framework, built upon a straightforward sequence-to-sequence learning approach. OFA achieves competitive performance in zero-shot learning, and is able to adapt to out-of-domain information without fine tuning [84].

SimVLM: SimVLM does not use auxiliary models such as Faster R-CNN [69] for region detection; instead, it uses only the language modeling loss on raw image inputs. Unlike other VLP approaches that require numerous pre-training stages and auxiliary objectives, it just requires one pass pre-training utilizing a single language modeling loss. All model parameters are pre-trained from scratch on noisy image-text data which helps in achieving zero-shot generalization. [86].

FLAMINGO: Pre-trained language models provide two-fold benefits, firstly they save the computation power and time and secondly, they store a large amount of knowledge in LM weights. Flamingo leverages these capabilities of pre-trained language models, to handle both open ended tasks such as VQA or image captioning and closed ended tasks such as classification. It accepts input as image or video and generates descriptive text. The authors leverage the power of pre-trained language models, to have strong generative language abilities. The architecture of Flamingo includes a vision encoder pre-trained with an approach similar to Contrastive Language-Image Pre-Training (CLIP) [35]. Vision encoder extracts the semantic and spatial features from the given input images or videos. To bridge the vision encoder and pre-trained large language model together, their weights are frozen and two learnable architectures perceiver-resampler are used [2].

3.2.3. Multimodal large language models

Multimodal large language models such as LLaVA [124], GPT-4V [125] and Gemini [126] are able to process visual and linguistic inputs parallelly, thus enabling them to handle complex tasks such as visual question answering in addition to image captioning. The key idea behind MLLMs lies in employing varied methodologies to integrate rich visual representations with generative capabilities of pre-trained Large Language Models (LLMs).

The most common methodology to enable text-only LLMs to process visual information is to use a lightweight ‘adapter’ module, which could be a simple linear layer or a multilayer perceptron (MLP). The main goal of this adapter module is to adapt visual representations from a visual encoder into the same embedding space as the LLM’s textual tokens. This key innovation requires fine-tuning only this small adapter module and the text-only LLMs achieves the capability to ‘see’ the image and generate linguistic output based on the visual input without changing the vast pre-trained knowledge base. LLaVA (Large Language and Vision Assistant) is based on this adapter based architectural paradigm. Alternative options explored by researchers include using convolutional layers in the adapter module.

One prominent drawback of using LLMs is resource-intensiveness since it requires processing a large number of visual tokens from image input before feeding it to LLMs. This bottleneck is mitigated by researchers in Q-Former Architecture (From BLIP-2) introduced by

[123]. The core idea is to use a lightweight transformer module whose task is to generate a small and fixed number of trainable tokens from the most salient visual features.

More complex and sophisticated MLLMs utilize dedicated modules to fuse visual and linguistic features dynamically moving beyond simpler projections by attention mechanisms. Based on this methodology mPLUG-owl [127] incorporated a ‘visual abstractor component’. Qwen-VL [128] used learnable queries along with a single-layer cross-attention mechanism. Other notable strategies involve integrating dense cross-attention blocks at several layers within pre-trained MLLM. These are often combined with a Perceiver model, which serves to reduce the number of visual tokens before they are incorporated into the main language model, further optimizing the fusion process.

Despite the efficiency of MLLMs there are certain limitations such as hallucination and faithfully visually grounded captions. Hallucination may result in subtle semantic errors such as inventing relationships or entire scene elements. There is also a need for new benchmarks evaluation frameworks that can accurately gauge the depth, semantic, and reasoning abilities of these advanced models.

3.2.4. Large vision-language models

In the early stages of VL model development for image captioning, architectures were primarily influenced by the encoder-decoder framework derived from Sequence-to-Sequence (seq2seq) learning. To optimize the utilization of visual features by the text decoder, various multimodal fusion methods were explored, often incorporating attention mechanisms. These attention mechanisms, including both inter-modality and intra-modality attention, were integrated into the encoder-decoder architecture to enhance performance. Inter-modality attention [53,34,91], implemented on the decoder side, facilitated the generation of captions based on specific image regions or concepts of interest, while intra-modality attention focused on incorporating semantic and spatial object relations into the visual encoder. These advancements played a pivotal role in improving the quality and accuracy of image captioning generated by early VL models. For instance, the authors in [95] introduced a novel approach by incorporating a Graph Convolutional Network (GCN) into the visual encoder. This integration allowed for the representation of both semantic and spatial relationships among objects within the image. Similarly, the authors of [30] proposed the use of an object relation Transformer, which utilizes geometric attention mechanisms to explicitly take into account the spatial relationships between input objects.

Large vision language models are capable of performing a wide array of tasks with zero-shot or few-shot prompting, making them multimodal general-purpose systems. Architecturally there are three main components of a LVLM. A vision encoder for which vision transformers (ViT) is a prominent choice among researchers, a pre-trained LLM like GPT or Llama [129] and vision-language connector module, which helps in alignment of visual representations with textual embeddings space. This module could be a simple linear projection layer to a more innovative and complex cross-attention module as in Q-Former in BLIP-2.

OFA [100] and PaLi [130] were some of the foundational models treating every task as a sequence generation problem using traditional encoder-decoder architecture. The input sequence is prefixed with task specific instructions thus eliminating the requirement of task-specific customizations. PaLi combined a large pretrained ViT with a language model like (UL2 [131]), the components were initially kept frozen and then the model was trained on varied multimodal tasks. Owing to its strong visual encoder PaLi performed well on tasks requiring nuanced visual understanding, such as optical character recognition (OCR). It processes images and text into a single sequence using which the language decoder generates outputs.

To make these architectures more efficient computationally as well as to make them more conversant, InstructBLIP [132] came with an innovative feature extraction module(Q-Former) and instruction tuning. Q-former extracts small and a fixed number of relevant visual features

based on the input text, instead of feeding a large number of visual tokens directly into LLM.

Qwen-VL series models, Qwen2-VL/Qwen2.5-VL from Alibaba cloud, have excellent fine-grained visual understanding as well as multilingual capabilities. Qwen2-VL [133] processes images of different resolutions dynamically, it also integrates Multimodal Rotary Position Embedding, effective in fusion of spatial context across multiple modalities. Despite impressive capabilities and performance there are certain key bottlenecks to be addressed such as hallucination which might generate incorrect details about a visual input, as well as robustness to adversarial attacks. Subtle changes in input phrasing might make these models brittle. (Table 4)

3.2.5. Contrastive learning

Researchers have also explored the self-supervised learning-based models for image captioning, which have shown state-of-the-art performance. Instead of relying on labeled data provided by humans, self-supervised learning algorithms generate their own supervision signal from the input data itself. Contrastive Language-Image Pre-training (CLIP) [35,42], is one such self-supervised learning approach which aims to understand the content and context of images in a way that facilitates generating relevant and coherent captions. CLIP operates by embedding the image and text pairs in a shared semantic space where related pairs (like an image and its description) are brought close together, while unrelated pairs are pushed apart. (Fig. 10) This is achieved through contrastive learning, a type of self-supervised learning that doesn't require human-labeled data but instead generates labels from the input itself. By comparing positive pairs with negative examples, CLIP learns to infer textual representations from unseen images, leveraging a large dataset for learning signals.

However, CLIP has limitations, including sensitivity to image quality, difficulty with abstract images, and challenges in handling ambiguity [80]. Various models have employed the CLIP framework which underlines the utility of CLIP in image-captioning tasks. [14,35], CLIP-Cap [60] model utilizes CLIP encoding to initiate the image captioning process. It employs a mapping network to integrate CLIP's semantic features into the caption generation pipeline. Subsequent fine-tuning with a language model like GPT-2 [68] refines these initial representations, yielding informative and contextually relevant captions. Leveraging CLIP's pre-existing understanding of visual and textual data, the model efficiently generates captions in a self-supervised manner. By training only, the mapping network while keeping CLIP and the language model frozen, the architecture achieves lighter, faster, and more resource-efficient performance while remaining competitive on benchmark datasets such as Conceptual Captions [70] and nocaps [11]. In [14] the authors have used the similarity scores generated by CLIP as rewards to provide feedback to the captioning model. By using the similarity scores as rewards, the captioning model can learn to generate captions that are more aligned with the content of the images.

3.2.6. Diffusion models

A significant shift in image captioning tasks is the emergence of diffusion models which serve as a promising generative tool. Diffusion models are able to capture subtle correlation between visual and textual

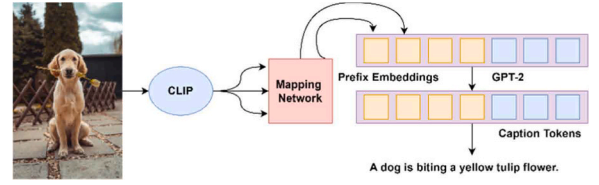


Fig. 10. Overview of CLIPCap Architecture [60].

elements better than generative adversarial networks (GANs) and autoregressive transformers. The key concept behind diffusion models is reversing a noise-injection process, this has two components; in the first stage, a Gaussian noise is incrementally added to the training data till a stage is reached where we are left with pure, unstructured noise. The primary task of the model is to learn a data distribution to reverse this process, it is trained to denoise the unstructured noisy data and reconstruct the original input.

At the time of inference, image captions are generated by applying this learned denoising process to random noise. Some of the methodologies utilized are, denoising diffusion probabilistic models (DDPMs) which employ Markov chains for both noising and denoising stages. Another approach is score-based generative models (SGMs) which focus on estimating the gradient of the data distribution at various noise levels. An efficient variant is Latent Diffusion Model (LDM) which performs the diffusion process within a compressed latent space using an autoencoder by removing pixel-level redundancies.

A standard diffusion model operates on continuous data whereas captions or text consists of discrete tokens. To mitigate this challenge, DDCap [134] model was developed, which addresses this challenge. It applies noise directly to text tokens converting them into a special mask token (Fig. 11). To predict the original caption, a transformer is trained to denoise the sequence.

In another approach the researchers used binary bits [137] to represent discrete caption tokens, which are further modeled as continuous analog values, thus even a standard diffusion which works on continuous data can be used. Leveraging the continuous diffusion, DiffCap [135] model embeds caption tokens into a continuous space and employs a standard Gaussian Diffusion process. A transformer model trained on CLIP-encoded image representations learns to denoise these continuous embeddings. This methodology produced more diverse captions than sequential models. LaDiC [136] model utilizes a split BERT architecture and a parallel decoding process to generate contextual captions accurately, overcoming the limitation such as slow inference and error propagation of autoregressive counterpart models. Certain drawbacks still persist in diffusion models which pose limitation to their real-time application such as resource-intensiveness owing to multi-step nature of denoising process, moreover since noise is added and removed gradually over multiple steps, this may corrupt the output quality as correct words generated in earlier steps may be erroneously changed in a later step. This requires careful tuning of the denoising schedule and model parameters.

Table 4

Experimental results on MSCOCO Image Captioning with cross entropy loss. The results are reported on the Karpathy test split.

Evaluation Metrics				
Model	BLEU-4	METEOR	CIDEr	SPICE
VL-T5 [14]	34.5	28.7	116.5	21.9
V-BART [14]	35.1	28.7	116.6	21.5
OSCAR [56]	34.5	29.1	115.6	21.9
OFA [100]	43.9	31.8	145.3	24.8
SimVLM [102]	40.6	33.7	143.3	25.4

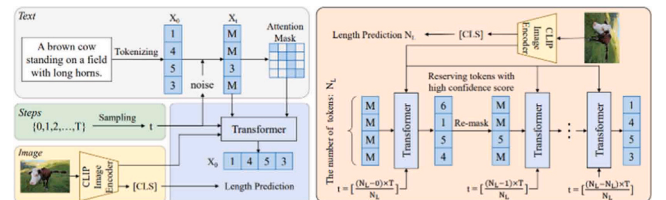


Fig. 11. Overview of DDCap Architecture (Process of training & inferences) [134].

3.3. Unsupervised and reinforcement learning

Note: Here we address RQ2. (What are the approaches in unsupervised, self-supervised and reinforcement learning?)

Many datasets consist primarily of unlabeled data and it is often impractical to accurately annotate huge quantities of unlabeled data. Unsupervised learning methods eliminate the dependency on high-quality image-caption pairs. Unsupervised and self-supervised learning techniques such as contrastive learning and adversarial learning are employed in image-captioning models [98]. Many unsupervised learning-based models use autoencoders and variational autoencoders due to their ability to learn meaningful representations of data through an encoder-decoder architecture, where the encoder compresses input data into a latent space representation, and the decoder reconstructs the original input from this representation. While autoencoders aim for a close approximation of input data, variational autoencoders incorporate probabilistic principles, allowing them to learn structured latent representations by modeling the distribution of latent variables.

An unsupervised learning method called an autoencoder (AE) trains a model to generate an output that closely resembles the input data (Fig. 12). Applications including data dimensionality reduction, image classification, object identification, and image denoising frequently make use of AEs. Three parts usually make up an AE's architecture: a latent space, a decoder, and an encoder. In order to capture important features and reduce information loss, the encoder component of the network compresses the input data into a lower-dimensional latent space representation. The input data is represented succinctly and meaningfully by the latent space. After that, the decoder uses this compressed form to reassemble the original input data. But conventional AEs are unable to provide completely new data samples because they are not explicitly designed for this purpose [87].

A Variational Autoencoder (VAE), on the other hand, is a unique kind of autoencoder that integrates ideas from variational Bayes inference [103]. The goal of VAEs, which were first introduced by the authors of [40] and later by the authors of [21], is to discover the training data's underlying distribution. Through sampling from this learned distribution, VAEs are able to produce new samples of data. The main distinction between VAEs and AEs is that the latter are meant to learn to represent the probability distribution of the data, whereas AEs concentrate on learning the compressed representation of input data and reconstructing it. By sampling from the learnt distribution, VAEs can produce fresh data samples, which makes them more suited for generative tasks than regular AEs [87].

Several cutting-edge methods for captioning images were developed which employ variational autoencoders to generate captions in an unsupervised manner. In [90,106], the authors present a novel structure for creating a variety of image captions that they name Conditional Variational Autoencoder (DCL-CVAE). Contrastive learning and sequential variational autoencoder are integrated in this system. In [85], another unsupervised learning approach which has shown state-of-the-art results in image captioning along with other vision-language tasks is Visual Commonsense R-CNN, the training objective of VC R-CNN differs fundamentally from other methods: it employs causal intervention instead of conventional likelihood. Various models leverage reinforcement learning for image captioning to improve the quality and coherence of generated captions. Within this framework, the authors in [82] examine the difficulty of crafting narrative paragraphs for a series

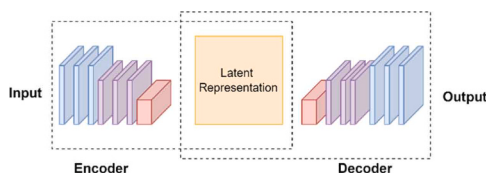


Fig. 12. A feed-forward deterministic autoencoder network.

of images, which is more demanding than generating captions for individual images because it requires well-organized narrative and emotive language. The difficulty of this task is increased by the limited training data. To overcome these obstacles, the authors suggest a sequence-to-sequence modeling strategy that is strengthened by adversarial training and reinforcement learning. As a story generator, they provide a hierarchical recurrent neural network that has been improved through reinforcement learning and rewards.

3.4. Stylized and controllable captioning

Note: Here we address RQ1. (Which image captioning models achieve superior performance for distinct caption types, including those requiring semantic and spatial alignment and emotional expression?)

A challenge in image captioning involves generating captions or image descriptions which are not only factually correct, but also capture the abstract idea behind the image, in particular, they must also capture the sentiment that an image may have, making them more human preferable [104]. In the same direction, the authors in [58] introduce SentiCap, a novel approach to image captioning by integrating sentiment analysis into the traditional CNN+RNN architecture. This model, composed of two parallel RNNs, adeptly captures both positive and negative sentiments within image descriptions. While one RNN serves as a background language model, the other specializes in sentiment representation, allowing for nuanced and emotionally resonant captions. A unique word-level regularizer further enhances the model's performance by emphasizing sentiment words during training, effectively blending the outputs of both RNN streams.

The authors in [27] address the task of Multi-Style Image Captioning (MSCap) using an adversarial learning network (Fig. 13). It seeks to develop a single model that can produce captions in different styles without the need for paired images. The standard image encoder is followed by four key modules that make up the framework: a style-dependent caption generator that generates a statement by using a style and an encoded image. A discriminator for captions that separates produced sentences from actual sentences. A Style classifier which identifies the style of the input sentence and a Back-translation module which ensures visually grounded stylized captions by enforcing cycle consistency between factual and stylized captions.

In a similar vein the authors in [101,120] introduce MemCap, which employs memory mechanisms to encode knowledge about linguistic styles, diverging from the prevalent reliance on language models. Its memory module stores embedding vectors representing phrases related to style, learned from the training corpus. To identify these phrases, MemCap employs a sentence decomposing algorithm that separates stylized sentences into parts reflecting style and content. During caption generation, the system extracts relevant style information from the memory module using attention mechanisms and integrates it into the language model.

Most of the captioning techniques often employ black-box architectures, which lack transparency and controllability from an external standpoint. Due to the infinite potential variations in image descriptions depending on the specific objectives and contextual nuances, a higher degree of controllability becomes necessary for captioning systems to

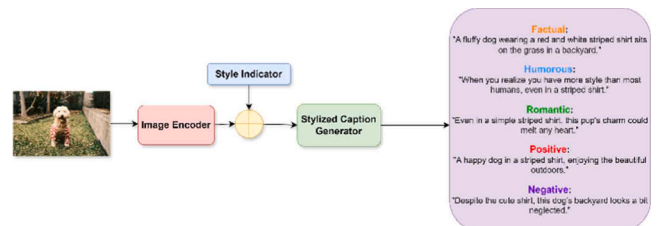


Fig. 13. Overview of a multi-style captioning model.

function well in complex settings [16]. Controllable captioning methodologies engage users in the image captioning process by soliciting their input to select and prioritize elements to be depicted. This user-provided guidance serves as a guiding agent for the caption generation process [73].

Recent advancements in captioning methodologies, as outlined in [16], facilitate the generation of diverse captions by integrating both grounding and controllability mechanisms. By leveraging control signals in the form of sequential instructions or sets of image regions, these methodologies employ recurrent architectures to generate captions that explicitly correlate with designated regions. This approach adheres to the constraints imposed by the provided control parameters, thereby enhancing the precision and adaptability of the generated captions [16].

Image captioning is typically done in English, but multilingual captioning tries to make it work in other languages too. There are two main ways to do this: one is to gather captions in various languages for popular datasets, like having Chinese and Japanese captions for COCO images or German captions for Flickr30K. The other way is to train systems that can handle multiple languages without needing paired captions in each language [73].

Upon analyzing Table 6, we note that for positive style captions, the MSCap [31] model achieves a higher CIDEr score on the SentiCap dataset, implying superior consensus and diversity in its generated captions. Conversely, the SentiCap model outperforms in METEOR score, indicating better fluency in its styled captions. For negative style captions SentiCap achieves both higher CIDEr and METEOR score. We examine perplexity and style classification accuracy for positive, negative, romantic, and humorous style captions. The Perplexity score measures how well a model predicts a sample. The lower the perplexity scores the better the performance, for the rest of the metrics the higher score is better. The style classification accuracy, refers to the accuracy achieved by a pre-trained style classifier in correctly identifying the style or category of a given caption. It is observed that MemCap demonstrates higher style classification accuracy compared to MSCap. This implies MemCap's superior ability to correctly identify the intended style of

captions.

Additionally, MemCap achieves lower perplexity scores across these styles. Lower perplexity scores signify reduced uncertainty in the language model's predictions, indicating higher fluency and coherence in the generated captions. Therefore, MemCap not only excels in accurately classifying different styles but also produces more fluent and coherent captions across various stylistic categories. To analyze the performance of various models in terms of semantic and spatial alignment we observe the SPICE scores in Table 7. It shows that SPICE score is higher for SimVLM and VinVL in comparison to other methods.

4. Applications

4.1. Remote-sensing

Remote sensing (RS) is a prominent research field and there has been significant research interest in remote sensing image captioning of the AI community. The predominant focus was on advancing the encoder-decoder architecture to better comprehend the complexities of aerial photography. In a consistent effort to improve feature extraction and obtain more descriptive and accurate captions transformer architecture was employed with integration of 'topic token' [138], which helps in guiding the generation process with global semantic information. Mask-guided training strategy was used to improve the diversity of captions as well as accuracy.

A significant innovation was combining transformers with reinforcement learning to achieve enhanced caption quality. The reinforcement learning component helped the model in adapting to smaller Remote sensing datasets. [139] Further scene attention model developed by [140] achieved the enhanced stability of captions by capturing enhanced global visual intelligence. Subsequently, [141] introduced multiscale methods that integrate scene-level and target-level information to generate enhanced image representations.

With the advent of Large Vision-language models, the latest research has been to develop LVLm based models for RS image analysis. One such novel approach is RS-LLaVA [142], which has been adapted particularly for remote-sensing captioning and visual question answering (VQA). Extensively pre-trained language models fine-tuned on aerial imagery helps in accomplishing multiple tasks (Fig. 14). To facilitate training and evaluation of multi-task capable models the authors also developed the RS-instructions dataset, which integrates multiple single-task datasets.

Despite these advancements, pertinent limitations exist, as noted by [139], there exists the risk of overfitting due to the relatively small size of publicly available RS captioning datasets compared to those for natural images. LVLms though improve the results but this comes at the cost of increased computations for training and inference. Although methods like low-rank adaptation are used to improve efficiency [142], resource intensity is a major drawback. Additionally, improving the diversity and specificity of generated captions remains an open problem, as models can tend to produce generic descriptions [138].

4.2. Underwater

Underwater image captioning is a significant task with applications in ocean geoscience and underwater remote-sensing. Research has shown that the performance of image captioning models depends on the pre-trained models however there is a significant bottleneck in using similar image captioning models trained on terrestrial ecosystems for interpreting underwater images. A better understanding of underwater scenes requires a holistic approach, the current image captioning models lack this broader context since they are based on encoding regional-features. A few other challenges are scale differences in underwater scenes and terrestrial scenes, inability to *recognize* visual elements like edges, shapes and outlines and lack of high-resolution large-scale dataset.

In order to resolve the issue of domain gap between natural images

Table 5

Summarization of popular datasets in image captioning along with their characteristics.

Dataset	Images	Captions per Image	Characteristics	Object Types
MS COCO	328,000	5	Everyday Scenes	91
Flickr30K Entities	31,783	5	Contains region-to-phrase correspondences	44,518
Visual Genome	108,077	50	Everyday scenes with objects, attributes and relationships	3.8 M
TextCaps	28,408	5	Consists of images with text present in them	-
VizWiz-Captions	39,181	5	Consists of captions of images taken by visually impaired people	-
Nocaps	15,100	10	Dataset for novel object captioning	500
FlickrStyle10k	10,000	1	Distinct Caption Style of humorous, poetic and factual (neutral)	4
SBU Captions	10,00,000	1	Single informal captions per image	-
ADE20K	20,000	-	Dense pixel level annotation for background and object	-
Visual Question Answering (VQA)	47,300	7	7 Ques-Answer pairs defining the context of background and object.	-

Table 6
Evaluation metrics of various multi-style image captioning models, based on four sentiments, positive, negative, romantic and humorous. MS COCO is used for factual captions, senticap for positive and negative sentiments and FlickrStyle 10k dataset is used for romantic and humorous sentiments.

	Sentiment					Evaluation Metric							
Model	Positive	Negative	Romantic	Humorous	Personality /Style	BLEU-4	CIDEr	SPICE	METEOR	ROUGE-L	Style classification accuracy (percent.)	Perplexity (lower is better)	Dataset
SentiCap [58]	Yes	-	-	-	-	10.8	54.4	-	16.8	36.5	-	-	SentiCap
SentiCap [58]	-	Yes	-	-	-	13.1	61.8	-	16.8	37.9	-	-	SentiCap
Generative model (UPDOWN with image encoder ResNeXt trained on 3.5 billion Instagram pictures) [71]	-	-	-	-	Traits selected from 215 possible personality traits such as dramatic, anxious, optimistic.	8	16.5	5.2	-	27.4	-	-	PERSONALITY-CAPTIONS
MSCap [27]	Yes	-	-	-	-	-	55.3	-	16.8	-	92.5	19.6	MSCOCO, SentiCap, FlickrStyle10K
MSCap [27]	-	Yes	-	-	-	-	51.6	-	16.2	-	93.4	19.2	MSCOCO, SentiCap, FlickrStyle10K
MSCap [27]	-	-	Yes	-	-	-	10.1	-	5.4	-	88.7	20.4	MSCOCO, SentiCap, FlickrStyle10K
MSCap [27]	-	-	-	Yes	-	-	15.2	-	5.3	-	91.3	22.7	MSCOCO, SentiCap, FlickrStyle10K
MemCap [101]	Yes	-	-	-	-	-	52.8	-	16.6	-	96.1	18.1	MSCOCO, SentiCap, FlickrStyle10K
MemCap [101]	-	Yes	-	-	-	-	59.4	-	15.7	-	98.9	18.9	MSCOCO, SentiCap, FlickrStyle10K
MemCap [101]	-	-	Yes	-	-	-	19.7	-	7.7	-	91.7	19.7	MSCOCO, SentiCap, FlickrStyle10K
MemCap [101]	-	-	-	Yes	-	-	18.5	-	7.2	-	97.1	17	MSCOCO, SentiCap, FlickrStyle10K

Table 7

Summarizing various image captioning models reviewed, listing the performance on key evaluation metrics on MSCOCO dataset with Karpathy test split.

Evaluation Metric					
Methods	BLEU-4	CIDEr	METEOR	ROUGE-L	SPICE
GIT [97]	-	148.8	-	-	-
CoCa [114]	40.9	143.6	33.9	-	24.7
CLIPCap [70]	33.53	113.08	27.45	-	21.05
VinVL [117]	41.0	140.9	31.1	-	25.4
Show and Tell [95]	32.1	99.8	25.7	-	-
Adaptive Attention [63]	33.2	108.5	26.6	-	-
Areas of Attention [77]	28.8	87.4	23.7	-	-
Up-down [5]	36.3	120.1	27.7	56.9	21.4
Object Relational Transformer [34]	38.6	128.3	28.7	58.4	22.6
X-linear attention [75]	39.7	132.8	29.5	59.1	23.4
Normalized and Geometry self-attention [32]	38.8	126.3	29.0	58.7	-
Context-aware attention [99]	35.8	111.3	27.8	56.4	-
Transformer and Knowledge Graph [119]	33.72	110.30	27.45	-	20.64
Unifying Vision and language via text-generation [14]	34.5	116.5	28.7	-	21.9
OFA [100]	43.9	145.3	31.8	-	24.8
SimVLM [102]	40.6	143.3	33.7	-	25.4
OSCAR [56]	36.58	124.12	30.4	-	23.17
AoANet [38]	38.9	129.8	29.2	58.8	22.4
Soft Attention (Show, attend and tell) [107]	24.3	-	23.90	-	-
Hard Attention (Show, attend and tell) [107]	25.0	-	23.04	-	-
ETA [55]	39.3	126.6	28.8	58.9	22.7
M2 [19]	39.1	131.2	29.2	58.6	22.6
DLCT (ResNext 101backbone) [65]	39.8	133.8	29.5	59.1	23.0

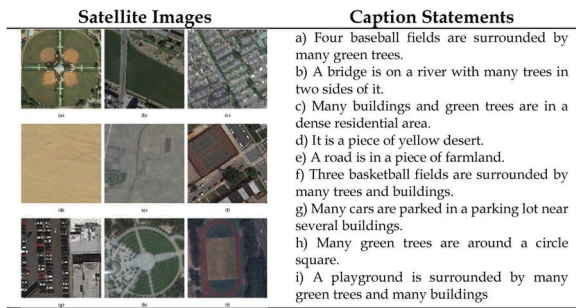


Fig. 14. Captions generated by Transformer+SCST model [139] on RSICD Dataset.

and underwater images a methodology was adopted by authors in which natural images were converted to “real-looking” degraded images. It was accomplished using physics-based submergence GAN [143]. The key idea behind submergence GAN is using three distinct generators (red, green and blue) with ResNet18 [46] backbone ensuring only color and intensity are changed while keeping the original scene intact. This methodology has been extended across six color domains to generate a large dataset using source images from ImageNet-100 [144].

Using this dataset, a ResNeXt [145] model is pre-trained with weights initialized from ImageNet dataset. In order to adapt to the low-resolution visual information and color characteristics the training focuses on optimizing the convolution layers. This training improves the model’s ability to handle underwater captioning considerably.

While this methodology is a promising leap forward in underwater image captioning, a potential drawback could be that the model may

struggle to perform on actual underwater images which might not be covered by six simulated color domains. The degraded images data generated might not capture the full spectrum of underwater complexities like non-uniform lighting, diverse water turbidity etc.

To address the limitations of current feature extraction models a novel model was developed Underwater Image Captioning model based on Scene-Object Feature Fusion (UICM-SOFF) [146]. The key idea is to use dual feature extraction, one for scenes and other for objects. Finally, a custom transformer-based encoder is used to fuse these features. These highly optimized scene-object features are inputted into a transformer decoder to generate highly relevant underwater image captions.

This model relies on running two separate computationally heavy networks for a single image, one for scene features and another for object features, making it resource-intensive as well as slowing the inference process thus making it unfeasible for redeployment on computationally resource limited platforms and real-world applications.

Quite opposite to the degraded image generation approach, authors of AquaSketch [147] developed a method to reduce the distortion along with enhancing details and background information like contours and background layers (Fig. 15). Firstly, shallow feature extraction and multiscale greyscale fusion is employed, later background sketch components are extracted like edges, vertices etc. These two components are fused to obtain an AquaDetail map. Swin-Transformer [148] branch is fed with this map and to obtain semantically rich and descriptive captions.

The underwater object information is improved using cross-scale fusion in a top-down manner, this is done to overcome the limitation posed by ignoring the multiscale features. Attention-based features fusion structure is employed to obtain cross-scale pyramid features. A significant innovation is the top-down fusion process as it enriches the fine-detail layers with essential context from the overall scene. This makes the model context-aware and preserves the subtle details of underwater scenes (Fig. 16).

While this model achieves significant results, potential modifications could be that the enhancement module could be improved by making it adaptive. The network could learn to apply different levels or types of enhancement based on the input image’s specific characteristics (e.g., turbidity, color distortion). This would allow it to preserve important environmental details while still improving the clarity of key objects.

4.3. Medical

If we look at the application domains of image captioning problem, most significant out of these is the medical or health care domain. Use of artificial intelligence in generating descriptive reports by looking at the medical images of various diseases is a crucial task which can be revolutionized by emerging technologies in the vision-language domain. Medical image captioning task is different from regular image captioning in the way that it highlights the relationship between image and clinical findings, also there are various medical image modalities such as radiography, MRI, ultrasound, computed tomography which have to be taken into consideration for generating detailed automated clinical reports. Medical image captioning can be used for diagnosis,

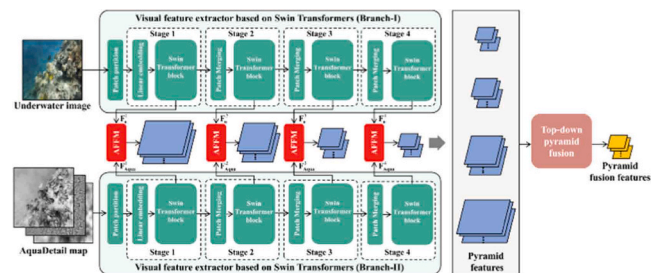


Fig. 15. AquaSketch Dual Branch Architecture [147].

Underwater Images	Reference Sentences
	A diver swims in the sea. In front of the diver is a coral reef. There are many small fish swimming in the coral reef. There are some coral reefs below the diver. A school of golden fish swims in the coral reef.
	A fish swimming in the sea. A fish swims from right to left in the sea. here is a columnar coral reef in front of the fish. Below the fish are a group of small coral reefs. There is a small fish swimming in the coral reef.

Fig. 16. Sample Annotation from UIEB Dataset [147].

treatment and report generation. For example, radiology report generation requires extensive expertise, hence reliable automatic report generation is much desirable.

There are certain bottlenecks in the medical image captioning task, first being the complexity of the task itself, since generating descriptive clinical reports from medical images is a challenging task, second is open access datasets with paired medical images and reports is sparse. Medical reports are descriptive in nature which are longer than typical image captions and they must be precise, reliable and must cover heterogeneous information, which may be difficult in case of rare diseases.

Various image captioning methods have been employed for the task of medical image captioning. Most recent vision-language pre-training paradigm has also been applied to the medical image captioning task. One of the main challenges in generating automatic medical reports from medical images is to detect abnormalities, since the training corpora has sparser abnormal cases. Producing thorough medical reports is a difficult task due to the length of medical reports. In order to resolve the first task of detecting patients' abnormalities Global label Pooling (GLP) is proposed [113]. Detecting several abnormalities is taken up as a multi-label classification problem. A CNN is used to generate feature maps and GLP mechanism is used to predict abnormality heat maps from feature maps.

In Global label pooling, a DenseNet except the last global pooling layer and fully connected layer is used and a feature map is obtained, using this feature map a label map is obtained using a convolutional layer. The label map indicates the heat maps for all the abnormalities. After detecting all possible abnormalities, a hierarchical RNN is used for caption prediction, viz-a-viz; a sentence RNN and word RNN. The sentence RNN responsible for generating sentences from image features determines the number of sentences needed for the resulting paragraph. It also generates a topic vector for each sentence. Then, using this topic vector, word RNN, predicts the words for each sentence individually. The outputs of this word-level RNN are then combined together to form the final paragraph.

Generating radiology reports is an intensive and time-consuming task, automatic image captioning is highly desirable in this domain, but generating detailed reports which highlight the linkage of a particular diagnosis with a region in the image is a challenging task, and it is made even more difficult by the absence of publicly available databases with paired medical images and reports, in order to tackle these challenges a generative encoder-decoder architecture was proposed in [115]. Their work mostly focuses on chest x-ray images (Fig. 17). By pre-training their image encoder on a large volume of chest x-ray images, the issue of a lack of datasets for this task was overcome, allowing it to correctly recognize 14 typical radiographic observations.

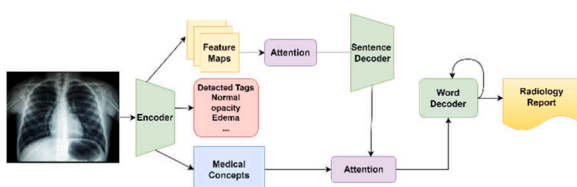


Fig. 17. A typical architecture of Medical Image Captioning Model

The image encoder was fed with multiple views of an image as a single input and in order to enforce consistency in predictions across different views Mean squared loss error was used. In addition to that, they also extracted most recurrent medical concepts from the radiology reports in the training data and used a concept classifier on top of image encoder.

Two hierarchical decoders were used in order to generate reports sentence by sentence. The two layers are composed of a word LSTM decoder that produces natural language output from sentence hidden states as input and a sentence LSTM decoder that generates sentence hidden states. Three different attention mechanisms are proposed to fuse Multiview image features; viz-a viz; concatenation of Multiview image features, early fusion of frontal and lateral global features with attention and late fusion of frontal and lateral global features with attention. Finally, the word LSTM decoder incorporates the medical concepts along with the output from LSTM sentence decoder and generates natural language output [115].

In another attempt to enhance medical image captioning the authors in [83] used the combined capability of seminal show, attend and tell (SAT) [107] model and generative pre-trained transformer [7] (GPT-3), the SAT model is used to identify the pathologies in the input images, with the help of an attention module, it creates a report based on the information found on x-ray. The GPT-3 model uses the SAT model's output as a seed. Using the MIMIC-CXR dataset [43], the refined GPT-3 [7] model produces a thorough report based on the pathologies identified by the SAT model.

Vision language pre-training paradigm has also been employed in the domain of medical image captioning. Researchers in [71] propose Medical Vision Language Learner (MedVILL). In this model a joint representation is learned from an image and its report and various vision-language understanding and generation downstream tasks are performed with fine tuning. After extracting visual information from the last convolution layer of the images, they flatten them along the spatial dimension using a CNN. BERT [21] is used for textual encoding and language feature embeddings are obtained, these two embeddings are then concatenated to input into the joint embedding component. Joint embedding component generates contextualized embeddings [71].

Note: In sections DATASETS and EVALUATION METRICS we attempt to answer RQ3. What are the primary datasets utilized for image captioning tasks, what evaluation metrics are commonly employed, how do these metrics assess caption quality, and what are the distinct characteristics of these datasets?

5. Datasets

5.1. General-purpose and specialized datasets

MS COCO dataset: widely recognized as one of the most prominent benchmarks for image captioning tasks. It contains a diverse collection of images, each annotated with multiple human-generated captions, capturing various objects, scenes, and activities in real-world contexts. With 2.5 million annotated object instances in 328k images across 91 distinct categories, MS COCO provides a comprehensive and diverse representation of visual content in real-world contexts. Each image is enriched with multiple annotations, including object segmentation masks, facilitating precise localization and recognition of objects within complex scenes [50].

Flickr30K entities dataset: Flickr30k Entities, an extension of the original Flickr30k dataset, enhances its utility by augmenting the existing 158,000 captions with 244,000 coreference chains. These chains create relationships between references to the same items in several descriptions for the same image, further enriched with 276,000 manually annotated bounding boxes. These annotations play a crucial role in advancing the field of automatic image description and grounded language understanding by facilitating tasks such as entity localization within images [67].

Visual genome dataset: In contrast to traditional datasets focused on perceptual tasks like image classification, Visual Genome emphasizes cognitive tasks such as describing images and answering questions. With the help of the deep annotations of objects, properties, and relationships provided by this dataset, models are better able to comprehend and make sense of the visual world. With over 108,000 images, each containing a wealth of object instances, attributes, and pairwise relationships, it is one of the densest and largest databases of image annotations and descriptions. Additionally, it has applications across dense image captioning and visual question answering tasks [44].

TextCaps: The TextCaps dataset addresses the need for image descriptions that encompass both visual content and accompanying written text, a crucial aid for visually impaired individuals in understanding their surroundings. Unlike previous approaches which primarily focus on automatically describing images and recognizing text separately, TextCaps integrates both aspects, presenting a novel challenge for models to comprehend text within the context of an image. With 145k captions corresponding to 28k images, this dataset requires models to not only recognize text but also to establish its relationship with visual elements and determine the pertinent information to include in the description. This task necessitates spatial, semantic, and visual reasoning, as models must navigate multiple text tokens and visual entities such as objects [72].

VizWiz: Captions- Comprised over 39,000 images sourced from visually impaired individuals, each paired with five captions, VizWiz-Captions offers a rich and diverse array of visual content [29].

Nocaps: The 'nocaps' benchmark includes 166,100 human-generated captions for 15,100 images from the *OpenImages* validation and test sets. It offers a strong basis for testing image captioning models on a wide range of visual concepts. By combining COCO image-caption pairs with *OpenImages* [43] image-level labels and object bounding boxes in the training dataset, 'nocaps' allows models to learn from diverse visual data, bridging the gap between training and real-world use. The inclusion of nearly 400 object classes in test images without enough training captions highlights the novelty and complexity of this benchmark [1].

FlickrStyle10k: This dataset consists of 10,000 images along with distinct styles such as humorous, poetic and factual (neutral) captions for each image.

Conceptual captions [155]: consist of 3.3 million caption pairs of images along with raw-descriptions with considerably varied style.

SBU captions dataset [158] consist of approximately 1 million image-caption pairs algorithmically derived from Flickr with user-generated titles and tags with mostly informal & single captions per image. The extensive dataset size makes it a pre-training captioning model's preferred benchmark.

ADE20K [156] comprises more than 20,000 images paired with dense pixel-level annotations for background elements as well as objects. Inherently by design it's not a captioning dataset but the availability of fine segmentations is really useful while training models to distinctively understand the object-background relationship and the descriptions are scene aware contextually.

Visual question answering (VQA) [157] also known as *Visual7W* is a fundamental captioning research source containing about 47,300 images. The images are provided with 7 question-answer pairs with specific focus on the object, its attributes and the relationship with the background context as well as image overall. The detailed information grounded with the context of the image is helpful in evaluating the model's performance while describing the fine details. (Table 5)

5.2. Remote-sensing datasets

Sydney-captions [151]: This dataset contains 613 highly detailed remote-sensing images, each cropped to 500×500 pixels with 0.5 m resolution. The dataset includes 7 classes of scenes with five unique captions for each image totaling to 3065 sentences. The images are

derived from the Google Earth platform.

UCM-captions [151]: This dataset contains 21 distinct scenes with 100 images per scene, each with a resolution of 256×256 and 0.3048 m resolution. These aerial images are derived from the UC Merced Land-Use dataset. The dataset contains a total of 10,500 captions with each image annotated with five captions.

5.3. Medical datasets

Radiology Object in Context version 2 (ROCOv2) [149]: This dataset is a medical multimodal dataset and an updated version of the ROCO dataset originally published in 2018, consisting of 79,789 images making it suitable for training image captioning models as well as for multi-label image classification using Unified Medical Language systems (UMLS). It contains radiological images along with medical captions and concepts derived from the PMC Open Access subset.

5.4. Underwater datasets

Underwater image enhancement benchmark (UIEB) [150]: Contains 950 real-world underwater images, corresponding to 890 images high-resolution reference images are available. The images cover a wide-variety of object categories ranging coral, fish, diving, marine life, wreckage etc. This dataset is suitable for validating the captioning model's performance in "into the wild" unknown scenarios.

6. Evaluation metrics

BLEU evaluates the machine-generated translation's quality. It offers a score depending on how similar a machine-generated translation is to one or more reference human translations. When comparing the machine generated translation to the reference translations, BLEU calculates how accurate the n-grams are. Precision is calculated for various values of n, usually up to 4. BLEU adjusts the precision score to account for situations in which the machine-generated translation may produce more n-grams than the reference translations. Each n-gram's count is capped at the most times it can appear in a single reference translation.

By comparing the lengths of the machine-generated and reference translations, it penalizes translations that are too short. This incentivizes the system to produce translations with lengths comparable to those of the reference translations. The final BLEU score is calculated by adding the updated precision scores for each n-gram and applying the brevity penalty. A perfect match between the machine-generated translation and the reference translation is indicated by a score of 1, which runs from 0 to 1. A notable drawback of BLEU is that it may not adequately account for variations in word choice, synonyms, or paraphrasing that could affect the overall quality of the translation [7,64]. We first compute the adjusted n-gram precision score on the test corpus before calculating the BLEU score [59].

We calculate the precision score based on the number of matched n-grams for each sentence in the test corpus. This entails calculating how many times each n-gram appears in the candidate translation and the reference translation. But we clip the counts—that is, we take the lowest count of each n-gram in both the reference translation and the candidate translation—in order to prevent overestimating precision because of n-gram repetition. These clipped numbers are added together for every sentence in the test corpus. We divide the sum of all candidate n-grams in all test corpus sentences by the sum of the clipped counts that were calculated.

Consider the modified precision score $P_n(X, Y)$ for candidate translations X and reference translations Y , given by:

$$P_n(X, Y) = \frac{\sum_i \sum_j \min \left\{ a_j(x_i), \max_{i \leq M} b_j(y_{ij}) \right\}}{\sum_i \sum_j \min \{ a_j(x_i) \}}$$

Where:

- x_i denotes the i^{th} candidate translation.
- y_{ij} denotes the j^{th} reference translation for the i^{th} candidate translation.
- $a_j(\cdot)$ represents the count of j -grams in candidate translations.
- $b_j(\cdot)$ represents the count of j -grams in reference translations.
- M denotes the maximum number of reference translations.

This formula computes the modified precision score, taking into account the minimum count between candidate j -grams and the maximum count of reference j -grams, divided by the minimum count of candidate j -grams.

Next, we calculate the *Brevity Penalty* (BP) which addresses the issue of penalizing candidate translations that are smaller than the length of reference translations, as follows;

$$BP = \begin{cases} 1 & \text{if } c_i > l \\ e^{\left(1 - \frac{l}{c_i}\right)} & \text{if } c_i \leq l \end{cases}$$

Where:

- BP denotes the Brevity Penalty.
- c_i represents the length of the candidate translation.
- l denotes the length of the reference corpus.

BLEU score is computed as,

$$BLEU = BP \times \exp\left(\sum_{n=1}^N \alpha_n \log p_n\right)$$

Where:

- BLEU represents the BLEU score.
- BP denotes the Brevity Penalty.
- p_n represents the modified n -gram precision.
- α_n denotes the positive weight for n -grams, summing to one.
- N represents the maximum n -gram length considered in the computation.

METEOR is predicated on the idea of unigram matching, which matches machine translation with reference human translations. METEOR matches words which originate from the identical stem as well as synonyms. If more than one reference translations are found, the best score is chosen after the translation scores are individually calculated against each reference [54,69].

Penalty Term and the harmonic mean of Precision and Recall (F_{mean}) can be represented as:

$$\text{Penalty} = \beta \cdot \left(\frac{\text{No. of Chunks}}{\text{No. of unigrams_matched}} \right)^\theta F_{mean}$$

$$= \frac{P_m \cdot R_m \cdot \alpha}{P_m \cdot \alpha + (1 - \alpha) \cdot R_m}$$

$$\text{Precision}(P_m) := \frac{P_m \cdot \frac{|m|}{\sum_k g_k(c_i)}}{\sum_k g_k(c_i)} \text{Recall}(R_m) : R_m = \frac{|m|}{\sum_k g_k(s_{ij})}$$

Where: β , θ , α are the hyperparameters set based on several datasets and $|m|$ represents the number of matched word pairs.

$\sum_k g_k(c_i)$ is the sum of all hypothesis words and $\sum_k g_k(s_{ij})$ is the sum of all reference words.

$$\text{METEOR} = (1 - \text{Penalty}) \cdot F_{mean}$$

The number of unigrams in the system translation that correspond to

the reference translation's unigram count is referred to as *unigrams mapped*.

Chunks: The smallest quantity of chunks that the matching unigrams can be grouped into. Every chunk is made up of neighboring system translation unigrams that are also mapped to neighboring reference translation unigrams.

ROUGE is a set of metrics used to evaluate the accuracy of machine-generated summaries. The metrics tally how many overlapping units—like word pairs, n -grams, and sequences of words—there are between the optimal human-written summaries and the computer-generated summary that needs to be assessed. The authors found that *ROUGE-S*, *ROUGE-L*, *ROUGE-W*, and *ROUGE-2* performed well in tasks involving the summarizing of a single document [58,69].

Rouge-L defines the longest common sub-sequence (LCS) as the longest identical fragment in the generated and reference sentences.

As examples, we consider the generated sentence G and the reference sentence R .

$$R_{lcs} = \frac{LCS(G, R)}{l_1}, \quad P_{lcs} = \frac{LCS(G, R)}{l_2} \text{ and } ROUGE-L = \frac{(1 + \beta^2) \cdot R_{lcs} \cdot P_{lcs}}{R_{lcs} + \beta^2 \cdot P_{lcs}}$$

where l_1 and l_2 represent the lengths of G and R respectively, and $\beta = \frac{P_{lcs}}{R_{lcs}}$. The only component to be taken into account in real-world applications is, $\beta \rightarrow \infty$, R_{lcs} . The precision without the F_{mean} function is the main focus of ROUGE.

CIDER evaluates the quality of image descriptions using human consensus. How frequently the n -grams found in candidate sentences are found in reference sentences is indicated by a consensus measure n -grams that are absent from the reference sentences but present in the candidate sentence may indicate divergence or inconsistency, hence they are penalized thereby leading to lower consensus score. Additionally, to account for the overall context of the dataset, common n -grams that occur frequently across all images are given lower weight in the consensus measure. These ubiquitous n -grams are likely to be less informative as they may represent generic language patterns rather than specific image-related content [69,93].

The average cosine similarity between the candidate phrase a_i and the reference sentences r_{ij} is used to calculate the $CIDER_n$ score for n -grams of length n . This score takes precision and recall into account.

$$CIDER_n(a_i, r_{ij}) = \frac{1}{m} \sum_j \frac{h_n(a_i) \cdot h_n(s_{ij})}{|h_n(a_i)| |h_n(r_{ij})|}$$

where $|h_n(a_i)|$ is the magnitude of the vector $h_n(a_i)$, and $h_n(r_{ij})$ is a vector produced by $h_k(a_i)$ corresponding to all n -grams of length n .

SPICE provides a distinctive approach in evaluating image captions by focusing on semantic content rather than n -gram overlap, unlike automatic image captioning metrics. The n -gram based strategy does not sufficiently reflect human assessments of quality. At its core SPICE works by calculating the F1-score between the scene graphs of human-generated caption and reference caption, this score gives the similarity of the semantic tuples (objects, attributes and relations).

This approach has achieved a system-level correlation of 0.88 with human-generated captions on MS COCO dataset outperforming CIDER (0.43) and METEOR (0.53) by a considerable margin. Additionally, the scene-graph helps in answering specific questions related to the model's understanding of colors and count more precisely. [3,69].

For *SPICE*, a caption Q is initially processed into a scene graph:

$$H(Q) = \langle O(Q), E(Q), K(Q) \rangle$$

where M denotes a set of object classes and $O(Q) \subseteq M$ is the set of objects referenced in Q , the set of edges representing the relationships between the objects is $E(Q) \subseteq O(Q) \times R \times O(Q)$.

A set of relation types is represented by R , and $K(Q) \subseteq O(Q) \times A$ is the collection of properties linked to the object.

A denotes attribute type sets.

Next, the function T produces a logical semantic tuple built as follows to assess how similar the candidate scene graph is to the reference one:

$$T(H(S)) \triangleq O(Q) \cup E(Q) \cup K(Q)$$

The binary matching operator $\otimes$ is used as a function to return matching tuples. Finally, the F-values of objects, attributes, and relations in the sentence are calculated to evaluate the accuracy:

$$P(Q, S) = \frac{|T(H(Q)) \otimes T(H(S))|}{|T(H(Q))|}$$

$$R(Q, S) = \frac{|T(H(Q)) \otimes T(H(S))|}{|T(H(S))|}$$

$$SPICE(Q, S) = F_1(Q, S) = \frac{2 \cdot P(Q, S) \cdot R(Q, S)}{P(Q, S) + R(Q, S)}$$

where $P(Q, S)$ and $R(Q, S)$ are precision and recall respectively.

ImageNet Sentence Similarity (IS): IS uses a pretrained model on ImageNet database to convert captions generated by an image captioning model and the human generated reference captions into high-dimensional vector embeddings. It evaluates the semantic similarity of these captions by analyzing the cosine similarity between these vector embeddings. Higher values of cosine similarity score are preferred as it signifies the two captions are semantically closer.

Self-CIDEr [154] as the name suggests evaluates how distinct a specific caption is as compared to other captions generated by the same model for the same image. This approach is utilized to encourage diversified captions and to avoid overly similar or redundant captions. At its core it uses CIDEr to evaluate output captions against other captions from the same image.

7. Future trends and challenges

7.1. Easier-to-train vision-language models

Vision-language models are not easier to train and their latent representation is complex. Hence future research trends will be towards developing medical vision-language models which are easier to train [52]. Since medical images are derived from diverse image modalities such as MRI scans, X-rays vision-language models are particularly useful in this scenario.

Feasibility: There's a push and wider interest of the AI community from massive and complex models towards building more efficient architectures (e.g., model pruning, quantization, knowledge distillation), hence the feasibility of developing more efficient and easier to train medical models is more prominent. An important consideration in this aspect is not to sacrifice the subtle details and understanding of medical images.

7.1.1. Implementation challenges

Performance trade-off: The primary bottleneck is the trade-off between model simplicity and diagnostic accuracy. Medical imaging requires the detection of subtle, low-signal features, which complex models are often better at capturing. Oversimplification of the model architectures could lead to a dangerous loss of clinical accuracy.

Benchmark disparity: Current academic benchmarks favor large models, making it difficult for researchers to get recognition for developing smaller, more efficient alternatives.

Hardware dependency: Training even "simpler" models still requires significant computational resources, which may not be available to all research institutions or hospitals.

7.2. Mitigating data scarcity (Few-Shot, Zero-Shot, and unsupervised learning)

Vision-language models require fine-tuning on sizable downstream datasets, in the medical domain where data is scarce, it hinders their capacity to accurately comprehend and generate insights from medical images and accompanying clinical text [61]. Few-shot and zero-shot models are required to tackle the problem of lack of data and to perform better on unseen inputs.

Feasibility: This is one of the most active and promising areas of research. The success of large foundation models pre-trained on web-scale data has shown that strong generalization is possible. Techniques like self-supervised learning, where models learn from the data itself without explicit labels, are already proving effective. Adapting these methods for the medical domain is highly feasible.

7.2.1. Implementation challenges

Clinical nuance: The biggest challenge is ensuring that self-supervised or few-shot learning captures nuanced clinical understanding, not just superficial visual patterns. A model might learn to associate a pattern with a disease without understanding the underlying pathology.

Negative Findings: A critical part of medical reporting is confirming the absence of disease. Zero-shot models, which are often trained to find what is there, struggle with reliably reporting what is not there.

Validation Complexity: Rare diseases pose a significant validation challenge for captioning models owing to a smaller sample size. This complexity can be resolved by curating expert-verified datasets for rare medical conditions.

7.3. Privacy preservation with federated learning

Image captioning of sensitive data such as medical data, where privacy is a main concern, can benefit from the integration of federated learning approaches.

Feasibility: The core technology for Federated Learning (FL) is mature, with established frameworks available (e.g., TensorFlow Federated). Given the strict privacy regulations in healthcare (like HIPAA), FL is not just feasible but is becoming a necessary approach for multi-institutional collaboration.

7.3.1. Implementation challenges

Data heterogeneity (Non-IID Data): This is the most significant technical challenge. Different hospitals use different scanners, imaging protocols, and see different patient demographics. When a model is trained on such varied data, the aggregated "global" model can perform poorly because the data distributions are not independent and identically distributed (non-IID).

Logistical and legal overhead: Establishing FL networks requires complex legal agreements, data-sharing protocols, and technical standardization across multiple institutions. This is often a slower process than the technology development itself.

Communication bottlenecks: The process involves sending model updates (gradients) over a network, which can be slow and resource-intensive, especially for large, state-of-the-art vision models.

7.4. Explainability (XAI)

With increasing applications of multimodal architectures in diverse fields and critical fields like healthcare and finance, there is a growing demand and interest in Explainability of such AI models. Just getting the answers or explanations is not enough; the model must be able to tell the "why" behind its reasoning. Hence the next future direction in this area would be towards building explainable models.

Feasibility: Basic Explainability methods, like heatmaps (e.g., Grad-CAM) that highlight where a model is "looking" in an image, are already

widely implemented and feasible. However, moving from these plausible visual aids to generating clinically faithful, textual explanations of the model's reasoning process is a much harder, ongoing research problem.

Implementation Challenges:

Faithfulness vs. plausibility: Many current XAI techniques produce explanations that are plausible (they look convincing to a human) but are not a faithful representation of the model's actual decision process. The explanation might highlight the correct area for the wrong reason.

Clinical Correlation vs. Causation: An explanation might highlight a feature that is statistically correlated with a disease but is not the actual pathology.

Workflow integration: The key challenge for adoption is presenting these explanations to clinicians in an intuitive, non-disruptive way within their existing software and workflow. If it takes too long to interpret the explanation, it may not be feasible.

8. Conclusion

Note: Here we answer RQ4, Considering the evaluation metrics employed, which image captioning models demonstrate superior performance?

In this literature survey, we have comprehensively covered the architectural evolution of the image captioning task. The main challenges of image captioning are to generate increasingly accurate, descriptive and context-aware captions. The key and foundational innovation in supervised learning-based approach to image captioning started with encoder-decoder architecture. There have been critical innovations and advancements in this paradigm to generate more semantically aligned captions; from global to more fine-grained regional features, the integration of scene graphs for relational understanding, and a significant paradigm shift brought by attention mechanisms.

The next inline highly performant architectures were early transformer-based models such as DLCT which demonstrated key enhancements on standard evaluation metrics. The recent paradigm shift in image captioning is towards self-supervised learning. This approach has enabled the development of large-scale Vision-Language Pre-trained (VLP) models, which have since evolved into even more powerful Multimodal Large Language Models (MLLMs) and Large Vision Language Models (LVLMs). Some top performers which utilize contrastive learning are OFA, GIT, and CoCa; there is immense promise in this approach with the emerging diffusion models. The survey also highlights advances in other learning strategies such as unsupervised and reinforcement learning. We have also explored specialized task-based captioning such as stylized captioning where models like MemCap have excelled.

Beyond the core architectures, we surveyed the practical application of these technologies in critical domains, highlighting their growing importance in remote sensing, underwater analysis, and medical imaging. This was supported by an analysis of the essential datasets that fuel innovation and the evaluation metrics used to measure progress. Finally, our review of future trends and challenges indicates that research is moving towards more scalable, efficient, and versatile models. The key opportunity lies in developing robust, domain-specific architectures that can overcome data scarcity and address the unique demands of real-world applications.

CRedit authorship contribution statement

sharma Aarti: Writing – review & editing, Writing – original draft, Conceptualization. **Hrishikesh Singh:** Writing – review & editing, Writing – original draft, Conceptualization. **Millie Pant:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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